A list of open problems in quantum information and quantum computation

"The formulation of a problem is often more essential than its solution, which may be merely a matter of mathematical or experimental skill."

— Albert Einstein and Leopold Infeld, *The Evolution of Physics* (1938)

Bikun Li

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Contents

1 Quantum capacity of a qubit Pauli channel **4**

2 Capacity-achieving codes for amplitude damping **5**

3 Distillable entanglement of Bell-diagonal states **6**

4 Achievability of the Rains bound under PPT-preserving channels **7**

5 Two-way quantum capacity: amplitude-damping channel **8**

6 Quantum LDPC codes at the Pauli hashing bound **9**

7 Entanglement cost of a qubit Bell-diagonal state **11**

8 Entanglement cost of an amplitude-damping-channel Choi state **13**

9 Three-dimensional self-correcting quantum memory **14**

10 NPT bound entanglement and the rank-five frontier **16**

11 Complete facet descriptions for Bell polytopes **18**

12 Relative entropy of entanglement for two qubits **19**

13 Long-range vacuum CHSH violation **20**

14 Tough error models **22**

15 Absolute separability from spectra **24**

16 Collective cost of tensor-power state preparation **26**

17 SIC-POVM existence in every dimension **27**

18 Weyl–Heisenberg-covariant SICs in every dimension **28**

19 Zauner-symmetric Weyl–Heisenberg SIC fiducials 29
20 Secret key from every entangled state 30
21 Lockability of two-way distillable entanglement 31
22 Honest-party lockability of distillable key 32
23 Quantum violations of bipartite Bell facets 33
24 Optimal CGLMP measurements for a maximally entangled state 34
25 Statistical strength of CGLMP measurements 35
26 Gaussian entanglement of formation beyond bisymmetry 36
27 One-bit simulation of partially entangled qubits 37
28 Universal simulation with two PR boxes 38
29 Finite-alphabet nonsignalling simulation of entangled qubits 39
30 Secret key from every Bell-nonlocal behavior 40
31 Semialgebraicity of closed quantum correlations 41
32 The PPT-squared conjecture 42
33 POVM steering threshold of higher-dimensional Werner states 43
34 Trace-exponential lower bound for matrix-word averages 45
35 Extensible causality and process purification 46
36 Constant trace-distance separability testing 47
37 Resources for implementing Gibbs-preserving channels 48
38 Polynomial shared-resource lower bounds for routing 49
39 Unconditional classical verification with one quantum prover 50
40 Existence of an eight-ququart perfect tensor 51
41 Real four-quhex absolutely maximally entangled state 52
42 Minimal-support frontier for absolutely maximally entangled states 53
43 Finite nontrivial LU moduli of AME states 55
44 Maximum number of mutually unbiased bases 56
45 Unclassified existence parameters for homogeneous AME states 57
46 Fixed-error parallel Stein lemma for quantum channels 58
47 Generalized Stein lemma for fully quantum channel resources 59

48 Square-root remainder in generalized quantum equipartition **60**

49 Quantum capacity of a bosonic thermal attenuator **62**

50 Positivity threshold for thermal-attenuator quantum capacity **64**

51 Energy-constrained quantum capacity of a thermal attenuator **65**

1 Quantum capacity of a qubit Pauli channel

Problem. What is the quantum capacity $\mathcal{Q}(\Lambda_{\mathbf{p}})$ of the qubit Pauli channel defined by

$$\Lambda_{\mathbf{p}}(\rho) = p_I\rho + p_X X\rho X + p_Y Y\rho Y + p_Z Z\rho Z, \qquad p_I + p_X + p_Y + p_Z = 1? \tag{1}$$

Equation (1) fixes the channel and the error-probability convention used throughout this section.

Status

Unsolved

Source

The exact-capacity question is implicit in the hashing lower bound of Bennett et al. and the strict degenerate-code improvement of DiVincenzo, Shor, and Smolin, which together leave the general Pauli-channel capacity undetermined [BDSW96], [DSS98].

Progress

- The hashing construction gives the achievable lower bound

$$\mathcal{Q}(\Lambda_{\mathbf{p}}) \geq \max\{0, 1 - H(\mathbf{p})\}, \qquad H(\mathbf{p}) := -\sum_{i\in\{I,X,Y,Z\}} p_i \log_2 p_i, \qquad 0\log_2 0 := 0. \tag{2}$$

  The quantity $1 - H(\mathbf{p})$ in Eq. (2) equals $I_c(I/2, \Lambda_{\mathbf{p}})$, the coherent information at the maximally mixed input (often called the symmetric coherent information), rather than the optimized one-shot coherent information in general [BDSW96], [Dev05].
- The coherent-information rate in Eq. (2) is achieved by efficient quantum polar codes for qubit Pauli channels. The first construction uses preshared entanglement (with zero consumption for sufficiently low noise), while a later construction removes the need for preshared entanglement [RDR12], [RSDR15].
- In the depolarizing case, write $p_I = f$ and $p_X = p_Y = p_Z = (1-f)/3$, and let $r := (1-f)/3$ be the per-Pauli error probability. Krohn-Grimberghe constructed an explicit rank-two, permutation-symmetric input across 45 channel uses and certified in exact rational arithmetic that its coherent information is positive at $r = 16239/250000 = 0.064956$. Monotonicity under channel post-processing therefore gives $\mathcal{Q}(\Lambda_{\mathbf{p}}) > 0$ whenever $f \geq 201283/250000 = 0.805132$. In particular, throughout $0.805132 \leq f < 0.81071$ the capacity is positive although the hashing expression in Eq. (2) is negative. Table 1 of the paper lists the preceding best printed positivity point as $r = 0.064657$, obtained from a floating-point eigensolver; the new point is machine-checkable but is not claimed to be the exact capacity threshold [KG26].
- For the qubit depolarizing subfamily, numerical optimization over channel extensions parameterized by the Stiefel manifold produces upper bounds on $\mathcal{Q}(\Lambda_{\mathbf{p}})$ that strictly improve the previously best flagged-extension bounds over the high-noise interval studied. The optimized bounds do not meet the known achievable rates, so they narrow the capacity gap without determining the capacity. The same work also proves that its amortized channel coherent information equals the ordinary one-shot channel coherent information for every channel, ruling out that amortization as a route to a stronger capacity lower bound [ZMFW25].

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[KG26] A. Krohn-Grimberghe, "A Certified Lower Bound on the Quantum-Capacity Threshold of the Depolarizing Channel," arXiv preprint (2026). arXiv:2608.15870v2.

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Comment

The exact quantum capacity remains unknown for a general qubit Pauli channel; the hashing rate is an achievable lower bound and need not be the capacity.

Tag

Pauli channels; Quantum capacity; Quantum Shannon theory; Coherent information; Qubit systems

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2 Capacity-achieving codes for amplitude damping

Problem. What is the constructive quantum code for achieving the quantum capacity $\mathcal{Q}(\mathcal{A}_p)$ of the qubit amplitude-damping channel

$$\mathcal{A}_p(\rho) = A_0\rho A_0^\dagger + A_1\rho A_1^\dagger, \qquad 0 \le p \le 1? \tag{1}$$

The operators appearing in Eq. (1) are

$$\begin{aligned} A_0 &= |0\rangle\langle 0| + \sqrt{1-p}\,|1\rangle\langle 1| = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-p} \end{pmatrix}, \\ A_1 &= \sqrt{p}\,|0\rangle\langle 1| = \begin{pmatrix} 0 & \sqrt{p} \\ 0 & 0 \end{pmatrix}. \end{aligned} \tag{2}$$

Equation (2) uses p as the decay probability of the excited state.

Status

Unsolved

Source

This constructive gap is implicit in the exact capacity formula of Wolf and Pérez-García and the polar-code construction of Wilde and Guha, which attains only the symmetric coherent-information rate for this channel [WPG07], [WG13].

Progress

- The channel is degradable for $0 \le p \le 1/2$ and antidegradable for $1/2 \le p \le 1$. Consequently,

$$\mathcal{Q}(\mathcal{A}_p) = \begin{cases} \max\limits_{0\le q\le 1} \big[h_2((1-p)q) - h_2(pq)\big], & 0 \le p \le \frac{1}{2}, \\ 0, & \frac{1}{2} \le p \le 1, \end{cases} \tag{3}$$

  where q is the excited-state population of the diagonal input and $h_2(x) := -x\log_2 x - (1-x)\log_2(1-x)$, with $0\log_2 0 := 0$. The capacity formula in Eq. (3) follows from the small-environment analysis [WPG07].

- For $0 \le p \le 1/2$, Wilde and Guha constructed a channel-adapted quantum polar code for degradable channels with a classical environment, explicitly including the amplitude-damping channel. It achieves

$$I_c(I/2, \mathcal{A}_p) = h_2\left(\frac{1-p}{2}\right) - h_2\left(\frac{p}{2}\right), \tag{4}$$

  the symmetric coherent-information rate. The rate in Eq. (4) is obtained with an efficient encoder and asymptotically vanishing entanglement consumption, although that work did not establish an efficient decoder [WG13]. Wilde and Renes subsequently gave a polar construction for arbitrary qubit-input channels that achieves the same rate using coherent successive-cancellation decoding [WR12].

References

[WPG07] M. M. Wolf and D. Pérez-García, "Quantum Capacities of Channels with Small Environment," *Physical Review A* **75**, 012303 (2007). doi:10.1103/PhysRevA.75.012303; arXiv:quant-ph/0607070.

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Comment

Because the amplitude-damping channel is nonunital, the maximizing population q in Eq. (3) is generally not 1/2. A code achieving only Eq. (4) therefore does not by itself achieve the full capacity. The unresolved constructive task is an explicit, efficient capacity-achieving scheme incorporating the required asymmetric input shaping.

Tag

Amplitude-damping channels; Quantum capacity; Quantum coding theory; Quantum polar codes; Degradable channels; Qubit systems

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3 Distillable entanglement of Bell-diagonal states

Problem. What is the distillable entanglement $D(\rho_{\mathbf{p}})$ under local operations and classical communication (LOCC) for the Bell-diagonal state

$$\rho_{\mathbf{p}} = p_I|\Phi^+\rangle\langle\Phi^+| + p_X|\Psi^+\rangle\langle\Psi^+| + p_Y|\Psi^-\rangle\langle\Psi^-| + p_Z|\Phi^-\rangle\langle\Phi^-|? \tag{1}$$

What is this LOCC protocol?

The Bell states in Eq. (1) are defined by

$$|\Phi^\pm\rangle := \frac{|00\rangle \pm |11\rangle}{\sqrt{2}}, \qquad |\Psi^\pm\rangle := \frac{|01\rangle \pm |10\rangle}{\sqrt{2}}. \tag{2}$$

Equation (2) fixes the phase convention. Assume $\sum_i p_i = 1$ and $p_I \geq 1/2 \geq p_i > 0$ for $i \in \{X, Y, Z\}$.

Status

Unsolved

Source

The problem is implicit in the gap between the Bell-diagonal distillation protocols of Bennett et al. and the PPT-based converse bounds introduced by Rains [BDSW96], [Rai99].

Progress

- The Rains bound gives

$$D(\rho_{\mathbf{p}}) \leq 1 - h_2(p_I), \qquad h_2(x) := -x\log_2 x - (1-x)\log_2(1-x). \tag{3}$$

  Equation (3) is the relevant upper bound for the present Bell-diagonal family [Rai99], [Rai01].
- The hashing protocol gives

$$D(\rho_{\mathbf{p}}) \geq \max\{0, 1 - H(\mathbf{p})\}, \qquad H(\mathbf{p}) := -\sum_i p_i \log_2 p_i. \tag{4}$$

  Recurrence followed by hashing improves on the direct use of Eq. (4) and shows that $D(\rho_{\mathbf{p}}) > 0$ whenever $p_I > 1/2$ [BBP+96], [BDSW96].
- Two-way protocols can strictly exceed the hashing rate in Eq. (4). Recurrence–hashing interpolation improves the rate for every full-rank entangled Bell-diagonal state, including the high-fidelity regime $p_I \to 1$ [VV05]. Adaptive protocols give related improvements for Werner/depolarizing states [HDM06], [AJSS26].

References

[Rai99] E. M. Rains, "An Improved Bound on Distillable Entanglement," *Physical Review A* **60**, 179–184 (1999). doi:10.1103/PhysRevA.60.179; arXiv:quant-ph/9809082.

[Rai01] E. M. Rains, "A Semidefinite Program for Distillable Entanglement," *IEEE Transactions on Information Theory* **47**, 2921–2933 (2001). doi:10.1109/18.959270; arXiv:quant-ph/0008047.

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[AJSS26] D. Abdelhadi, T. Jochym-O'Connor, V. Siddhu, and J. Smolin, "Adaptive Channel Reshaping for Improved Entanglement Distillation," *Physical Review Research* **8**, 013018 (2026). doi:10.1103/8kdf-37gp; arXiv:2410.22295.

Comment

The exact LOCC distillable entanglement is not known for a general full-rank Bell-diagonal state; the gap between constructive lower bounds and the Rains upper bound remains the central question.

Tag

Bell-diagonal states; Entanglement distillation; Local operations and classical communication; Entanglement measures; Rains bound; Qubit systems

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4 Achievability of the Rains bound under PPT-preserving channels

Problem. Consider distilling the bipartite Bell-diagonal state

$$\rho_{\mathbf{p}} = p_I|\Phi^+\rangle\langle\Phi^+| + p_X|\Psi^+\rangle\langle\Psi^+| + p_Y|\Psi^-\rangle\langle\Psi^-| + p_Z|\Phi^-\rangle\langle\Phi^-|, \tag{1}$$

where $p_i > 0$ and $p_I + p_X + p_Y + p_Z = 1$, and $|\Phi^\pm\rangle := (|00\rangle \pm |11\rangle)/\sqrt{2}$ and $|\Psi^\pm\rangle := (|01\rangle \pm |10\rangle)/\sqrt{2}$. Is the Rains bound of the state in Eq. (1) achievable by a positive-partial-transpose-state-preserving (PPT-state-preserving, or PPT-preserving) quantum channel? If so, what is the constructive quantum channel? A channel is PPT-state-preserving if every PPT input state is mapped to a PPT output state.

Status

Unsolved

Source

Rains introduced the PPT-preserving distillation framework and its semidefinite-programming bound; the present Bell-diagonal achievability question is an implicit specialization of that work [Rai99], [Rai01].

Progress

- Rains showed that the Rains bound $R(\rho_{\mathbf{p}})$ upper-bounds the entanglement distillable from $\rho_{\mathbf{p}}$ by PPT-preserving operations:

  $$D_{\mathrm{PPT}}(\rho_{\mathbf{p}}) \leq R(\rho_{\mathbf{p}}). \tag{2}$$

  Equation (2) is the relevant converse bound for the operational class in this problem [Rai99], [Rai01].

References

[Rai99] E. M. Rains, "An Improved Bound on Distillable Entanglement," *Physical Review A* **60**, 179–184 (1999). doi:10.1103/PhysRevA.60.179; arXiv:quant-ph/9809082.

[Rai01] E. M. Rains, "A Semidefinite Program for Distillable Entanglement," *IEEE Transactions on Information Theory* **47**, 2921–2933 (2001). doi:10.1109/18.959270; arXiv:quant-ph/0008047.

Comment

The problem asks whether the upper bound in Eq. (2) is achievable for the Bell-diagonal family in Eq. (1), and, if so, for an explicit construction of a PPT-preserving channel attaining it.

Tag

Bell-diagonal states; Rains bound; PPT-preserving operations; Entanglement distillation; Semidefinite programming; Entanglement measures

ID

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5 Two-way quantum capacity: amplitude-damping channel

Problem. What is the two-way quantum capacity $\mathcal{Q}_2(\mathcal{A}_p)$ of the qubit amplitude-damping channel

$$\mathcal{A}_p(\rho) = A_0 \rho A_0^\dagger + A_1 \rho A_1^\dagger, \qquad 0 \le p \le 1? \tag{1}$$

The operators in Eq. (1) are

$$\begin{aligned} A_0 &= |0\rangle\langle 0| + \sqrt{1-p}\,|1\rangle\langle 1| = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-p} \end{pmatrix}, \\ A_1 &= \sqrt{p}\,|0\rangle\langle 1| = \begin{pmatrix} 0 & \sqrt{p} \\ 0 & 0 \end{pmatrix}. \end{aligned} \tag{2}$$

Equation (2) uses p as the decay probability of the excited state. Here $\mathcal{Q}_2$ permits adaptive local operations and unlimited two-way classical communication between uses of the channel.

Status

Unsolved

Source

The question is implicit in the nonmatching achievable and converse bounds for the amplitude-damping channel reported by Pirandola, Laurenza, Ottaviani, and Banchi [PLOB17].

Progress

- Pirandola, Laurenza, Ottaviani, and Banchi bounded the two-way quantum capacity using an achievable reverse-coherent-information rate from below and the channel's squashed entanglement from above:

$$\max_{0 \le u \le 1} \left[h_2(u) - h_2(pu) \right] \le \mathcal{Q}_2(\mathcal{A}_p) \le h_2\left(\frac{1}{2} - \frac{p}{4} \right) - h_2\left(1 - \frac{p}{4} \right), \tag{3}$$

  where $h_2(x) := -x \log_2 x - (1-x)\log_2(1-x)$, with $0 \log_2 0 := 0$. The lower rate in Eq. (3) is achievable with a final round of backward classical communication, whereas the upper rate follows from a balanced amplitude-damping squashing channel [PLOB17].

References

[PLOB17] S. Pirandola, R. Laurenza, C. Ottaviani, and L. Banchi, "Fundamental Limits of Repeaterless Quantum Communications," *Nature Communications* **8**, 15043 (2017). doi:10.1038/ncomms15043; arXiv:1510.08863.

Comment

The bounds in Eq. (3) do not coincide in general. Determining $\mathcal{Q}_2(\mathcal{A}_p)$ therefore requires either an improved two-way-assisted protocol, a tighter converse bound, or both. This problem concerns the same amplitude-damping channel as Problem 2, but Problem 2 asks for a constructive code achieving the unassisted quantum capacity $\mathcal{Q}(\mathcal{A}_p)$, whereas the present problem asks for the two-way quantum capacity $\mathcal{Q}_2(\mathcal{A}_p)$.

Tag

Amplitude-damping channels; Two-way quantum capacity; Quantum Shannon theory; Quantum communication; Qubit systems

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6 Quantum LDPC codes at the Pauli hashing bound

Problem. For a probability vector $\mathbf{p} = (p_I, p_X, p_Y, p_Z)$, the qubit Pauli channel is

$$\Lambda_{\mathbf{p}}(\rho) = p_I\rho + p_X X\rho X + p_Y Y\rho Y + p_Z Z\rho Z, \qquad p_I + p_X + p_Y + p_Z = 1, \tag{1}$$

where every $p_i \geq 0$, I is the identity, and X, Y, Z are the Pauli operators. The qubit depolarizing channel is the specialization of Eq. (1) given by

$$\mathcal{D}_p(\rho) = (1-p)\rho + \frac{p}{3}\big(X\rho X + Y\rho Y + Z\rho Z\big), \qquad 0 \leq p \leq 1. \tag{2}$$

Thus Eq. (2) has $\mathbf{p} = (1-p, p/3, p/3, p/3)$. For the general channel in Eq. (1), define the hashing rate by

$$R_{\text{hash}}(\mathbf{p}) := \max\{0, 1 - H(\mathbf{p})\}, \qquad H(\mathbf{p}) := -\sum_{i\in\{I,X,Y,Z\}} p_i \log_2 p_i, \qquad 0\log_2 0 := 0. \tag{3}$$

A family of stabilizer codes $[[n, k_n]]$ is quantum low-density parity-check (LDPC) if it has stabilizer generators of uniformly bounded weight and each qubit participates in a uniformly bounded number of generators. Does there exist, for a given $\mathbf{p}$, such a family with decoding error tending to zero under $\Lambda_{\mathbf{p}}^{\otimes n}$ and asymptotic transmission rate $R := \liminf_{n\to\infty} k_n/n$ satisfying $R \geq R_{\text{hash}}(\mathbf{p})$ from Eq. (3)?

Status

Unsolved

Source

Contributor: Bikun Li.

Progress

- The XZZX surface code is an explicit quantum LDPC family whose numerically estimated code-capacity threshold closely matches the zero-rate hashing threshold for every single-qubit Pauli channel; for depolarizing noise its reported threshold is 18.7(1)%. It encodes only $O(1)$ qubits into $n = O(d^2)$ qubits, however, so its asymptotic rate is zero [BAT+21].
- Kasai reported a nonbinary quantum LDPC code of rate 1/3 with 312,000 physical and 104,000 logical qubits, attaining frame-error rate 10^{-4} at depolarizing error probability $p = 9.45\%$. This is strong finite-length numerical progress, but at that p Eq. (3) gives $R_{\text{hash}} \simeq 0.399 > 1/3$, and no asymptotic capacity-achieving theorem is proved [Kas25].
- More recently, fixed-degree quantum LDPC ensembles were proved to have nonvanishing rate and relative distance with high probability; selected finite degree choices attain the CSS Gilbert–Varshamov distance bound. These distance guarantees do not by themselves establish reliable Pauli channel transmission at the rate in Eq. (3) [Kas26].
- Quantum polar codes attain the hashing rate without establishing the LDPC constraints. The entanglement-assisted CSS construction of Renes, Dupuis, and Renner has net rate $1 - H(\mathbf{p})$ and efficient operations for Pauli channels, but can consume preshared ebits [RDR12]. The later unassisted two-level CSS construction has vanishing error and rate at least $\max\{0, 1 - H(\mathbf{p})\}$, with $O(n \log n)$ encoding and decoding once its frozen sets are specified; efficient construction of the outer frozen set was left open [RSDR15]. Neither work proves bounded check weight or bounded qubit degree.

References

[BAT+21] J. P. Bonilla Ataides, D. K. Tuckett, S. D. Bartlett, S. T. Flammia, and B. J. Brown, "The XZZX Surface Code," *Nature Communications* **12**, 2172 (2021). doi:10.1038/s41467-021-22274-1; arXiv:2009.07851.

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[RSDR15] J. M. Renes, D. Sutter, F. Dupuis, and R. Renner, "Efficient Quantum Polar Codes Requiring No Preshared Entanglement," *IEEE Transactions on Information Theory* **61**, 6395–6414 (2015). doi:10.1109/TIT.2015.2468084; arXiv:1307.1136.

Comment

Polar codes attain the hashing rate without LDPC constraints; the other results separately address threshold, finite-block performance, or distance. The open task is a quantum LDPC theorem for Eq. (1) with vanishing decoding error and rate at least Eq. (3).

Tag

Quantum LDPC codes; Stabilizer codes; Pauli channels; Depolarizing channels; Hashing bound; Decoding algorithms

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7 Entanglement cost of a qubit Bell-diagonal state

Problem. What is the entanglement cost of a qubit Bell-diagonal state for an arbitrary probability vector $\mathbf{p} = (p_I, p_X, p_Y, p_Z)$? Let $|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$ and $|\Phi_P\rangle = (I \otimes P)|\Phi^+\rangle$ for $P \in \{I, X, Y, Z\}$. The state is

$$\rho_{\mathbf{p}} := \sum_{P\in\{I,X,Y,Z\}} p_P |\Phi_P\rangle\langle\Phi_P|, \qquad p_P \geq 0, \qquad \sum_{P\in\{I,X,Y,Z\}} p_P = 1. \tag{1}$$

For the state in Eq. (1), determine $E_C(\rho_{\mathbf{p}})$ for every $\mathbf{p}$ in the probability simplex, where E_C is the infimum asymptotic rate of ebits consumed by LOCC protocols that prepare $\rho_{\mathbf{p}}^{\otimes n}$ with trace-norm error tending to zero.

Status

Unsolved

Source

The problem is implicit in the regularized-entanglement-of-formation characterization of entanglement cost and Wootters' one-copy formula for two-qubit states [HHT01], [Woo98].

Progress

- Let $p_\star := \max\{p_I, p_X, p_Y, p_Z\}$, $C_{\mathbf{p}} := \max\{0, 2p_\star - 1\}$, and $h_2(x) := -x \log_2 x - (1-x)\log_2(1-x)$, with $0 \log_2 0 := 0$. Regularized relative entropy of entanglement, entanglement cost, and Wootters' one-copy entanglement of formation give the rigorous bounds

$$L(p_\star) = E_R^\infty(\rho_{\mathbf{p}}) \leq E_C(\rho_{\mathbf{p}}) \leq E_F(\rho_{\mathbf{p}}) = h_2\left(\frac{1+\sqrt{1-C_{\mathbf{p}}^2}}{2}\right), \qquad L(t) := \begin{cases} 0, & 0 \leq t \leq \frac{1}{2}, \\ 1 - h_2(t), & \frac{1}{2} < t \leq 1. \end{cases} \tag{2}$$

  Here $E_C = E_F^\infty$ [HHT01]; Wootters gives the right endpoint [Woo98]; and additivity of the relative entropy of entanglement gives $E_R^\infty(\rho_{\mathbf{p}}) = L(p_\star)$ [ZCH10]. Thus Eq. (2) solves $p_\star \leq 1/2$ but generally leaves a gap for $p_\star > 1/2$. A recent semidefinite-programming construction gives a faithful, efficiently computable lower bound on $E_C(\rho_{\mathbf{p}})$ for every entangled two-qubit Bell-diagonal state, but does not close the gap [WJZ25].

- If at most two Bell probabilities are nonzero, say q and $1-q$, then the entanglement of formation is strongly additive and

$$E_C(\rho_{\mathbf{p}}) = E_F(\rho_{\mathbf{p}}) = h_2\left(\frac{1}{2} + \sqrt{q(1-q)}\right). \tag{3}$$

  Vidal, Dür, and Cirac proved Eq. (3) for a mixture of $|\Phi^+\rangle$ and $|\Phi^-\rangle$; local unitaries extend it to any pair of Bell states [VDC02]. This settles the rank-at-most-two boundary of the family, not generic rank-three or rank-four states.

- On the isotropic line singled out in the question, $p_X = p_Y = p_Z$, write the probabilities and the known one-copy value as

$$p_I = F, \qquad p_X = p_Y = p_Z = \frac{1-F}{3}, \qquad E_F(\rho_{\mathbf{p}}) = \begin{cases} 0, & 0 \leq F \leq \frac{1}{2}, \\ h_2\left(\frac{1}{2} + \sqrt{F(1-F)}\right), & \frac{1}{2} < F \leq 1. \end{cases} \tag{4}$$

  Terhal and Vollbrecht determined this one-copy quantity [TV00]; it is also the qubit specialization of Wootters' formula. For $1/2 < F < 1$, no proof is known that it equals the asymptotic LOCC entanglement cost.

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Comment

Equation (4) determines E_C only for $F \leq 1/2$ and $F = 1$; for $1/2 < F < 1$ it is an E_F formula. The open task is to regularize E_F for entangled rank-three and rank-four Bell-diagonal states. Problem 8 asks the analogous question for a different two-qubit family.

Tag

Bell-diagonal states; Entanglement cost; Entanglement measures; Local operations and classical communication; Qubit systems

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8 Entanglement cost of an amplitude-damping-channel Choi state

Problem. What is the entanglement cost of the Choi state of the qubit amplitude-damping channel

$$\mathcal{A}_p(\rho) = A_0\rho A_0^\dagger + A_1\rho A_1^\dagger, \qquad 0 \le p \le 1? \tag{1}$$

The Kraus operators in Eq. (1) are

$$\begin{aligned} A_0 &= |0\rangle\langle 0| + \sqrt{1-p}\,|1\rangle\langle 1| = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-p} \end{pmatrix}, \\ A_1 &= \sqrt{p}\,|0\rangle\langle 1| = \begin{pmatrix} 0 & \sqrt{p} \\ 0 & 0 \end{pmatrix}. \end{aligned} \tag{2}$$

In Eq. (2), p is the decay probability of the excited state. Let $|\Phi^+\rangle_{RA} = (|00\rangle + |11\rangle)/\sqrt{2}$. The normalized Choi state of the channel in Eq. (1) is

$$\begin{aligned} \omega_p^{RB} &:= (\mathrm{id}_R \otimes \mathcal{A}_p)(|\Phi^+\rangle\langle\Phi^+|_{RA}) \\ &= \frac{1}{2}\Big[|00\rangle\langle 00| + \sqrt{1-p}\big(|00\rangle\langle 11| + |11\rangle\langle 00|\big) + (1-p)|11\rangle\langle 11| + p|10\rangle\langle 10|\Big]. \end{aligned} \tag{3}$$

Here the first and second entries in each ket in Eq. (3) label R and B, respectively. Thus the question is to determine $E_C(\omega_p)$, the asymptotic number of ebits per copy required to prepare many copies of ω_p by local operations and classical communication.

Status

Unsolved

Source

The question is implicit in the identity between entanglement cost and regularized entanglement of formation, together with Wootters' single-copy formula applied to this Choi state [HHT01], [Woo98].

Progress

- Wootters' two-qubit formula, together with the concurrence of ω_p, gives the exact single-copy entanglement of formation

$$C(\omega_p) = \sqrt{1-p}, \qquad E_F(\omega_p) = h_2\left(\frac{1+\sqrt{p}}{2}\right), \tag{4}$$

  where $h_2(x) := -x\log_2 x - (1-x)\log_2(1-x)$, with $0\log_2 0 := 0$, [Woo98]. Equation (4) determines one copy exactly, but it does not by itself determine the asymptotic entanglement cost.
- The entanglement cost equals the regularized entanglement of formation

$$E_C(\omega_p) = \lim_{n\to\infty} \frac{1}{n} E_F(\omega_p^{\otimes n}) \le E_F(\omega_p) = h_2\left(\frac{1+\sqrt{p}}{2}\right) \tag{5}$$

  [HHT01]. Consequently, Eq. (5) reduces the problem to evaluating the regularization for this particular family of Choi states.

References

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Comment

The unresolved step is to decide whether regularization lowers the single-copy value in Eq. (4); equivalently, one must evaluate the entanglement of formation on tensor powers of this Choi state. Problems 2 and 5 concern the same channel family but ask about channel capacities, whereas the present problem asks about the entanglement cost of a bipartite state associated with the channel.

Tag

Amplitude-damping channels; Choi states; Entanglement cost; Entanglement measures; Local operations and classical communication; Qubit systems

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9 Three-dimensional self-correcting quantum memory

Problem. Does there exist a passive self-correcting quantum memory in three spatial dimensions? Consider finite-dimensional spins on a three-dimensional lattice Λ_L of linear size L and a Hamiltonian $H_L = \sum_{X \subseteq \Lambda_L} h_X$ whose interaction range, local strength $\|h_X\|$, and number of terms incident on each spin are bounded independently of L. Let its ground space $\mathcal{C}_L$ encode a logical space $\mathcal{Q}_L$ with $\dim \mathcal{Q}_L \geq 2$, and let $\mathcal{V}_L : \mathcal{S}(\mathcal{Q}_L) \to \mathcal{S}(\mathcal{C}_L)$ be the channel induced by an isometric encoding into $\mathcal{C}_L$.

During storage no active control or error correction is permitted. The encoded state evolves under a local thermal channel $\mathcal{E}_{t,\beta}^{(L)}$ at inverse temperature β, such as a Davies semigroup, followed only at readout by a decoder $\mathcal{D}_L$. For a fixed $0 < \varepsilon < 1$, define the worst-case storage time by

$$\tau_L(\beta, \varepsilon) := \sup \left\{ t \geq 0 : \sup_{0 \leq s \leq t} \sup_{\rho \in \mathcal{S}(\mathcal{Q}_L)} \left\| \left(\mathcal{D}_L \circ \mathcal{E}_{s,\beta}^{(L)} \circ \mathcal{V}_L \right)(\rho) - \rho \right\|_1 \leq \varepsilon \right\}, \tag{1}$$

where $\mathcal{S}(\mathcal{Q}_L)$ is the set of logical states. The problem is to construct such a Hamiltonian family with efficient decoders and a finite critical inverse temperature β_c for which

$$\beta > \beta_c \quad \Longrightarrow \quad \lim_{L \to \infty} \tau_L(\beta, \varepsilon) = \infty. \tag{2}$$

Equation (2) requires the lifetime in Eq. (1) to diverge at fixed nonzero temperature as $L \to \infty$; growth that terminates at a temperature-dependent system size is only partial self-correction.

Status

Unsolved

Source

Brown et al. explicitly identify the construction of a self-correcting quantum memory in three spatial dimensions as an outstanding open problem [BLP+16].

Progress

- Self-correcting quantum memories are known from four-dimensional local models, but the existence of a three-dimensional example satisfying Eq. (2) remains open. This status and the standard thermal-memory framework are summarized by Brown *et al.* and by the Error Correction Zoo [BLP+16], [AF26].
- Haah's three-dimensional cubic code has no string-like logical operators and possesses a logarithmic energy barrier. Under Davies dynamics, Bravyi and Haah proved a low-temperature lifetime bound of the form

$$T_{\mathrm{mem}} \geq L^{c\beta} \quad \text{for some } c > 0 \text{, but only when} \quad L \leq L^\star \sim e^{\beta/3}. \tag{3}$$

  Because the admissible size in Eq. (3) is bounded at fixed β, the cubic code is partially self-correcting but does not satisfy Eq. (2) [BH13].
- Recent three-dimensional layer codes have polynomial energy barriers and provably exhibit partial self-correction. For suitable families, their memory time grows exponentially with L only up to a temperature-dependent cutoff that is exponential in β; they still fall short of strict self-correction in the fixed-temperature thermodynamic limit [Wil25].

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Comment

The central gap is between a lifetime that grows with L only below a temperature-dependent size cutoff and the uniform thermodynamic-limit divergence required by Eq. (2). A complete solution must either construct a genuinely three-dimensional local model with the latter behavior or prove an appropriately general three-dimensional no-go theorem.

Tag

Self-correcting quantum memories; Quantum memories; Open quantum systems; Many-body quantum systems; Quantum thermodynamics; Topological quantum codes

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10 NPT bound entanglement and the rank-five frontier

Problem. Does there exist a finite-dimensional bipartite state with non-positive partial transpose (NPT) that is undistillable by local operations and classical communication (LOCC)? For a density operator ρ_{AB}, partial transposition on B is denoted by $\rho_{AB}^{T_B} := (\mathrm{id}_A \otimes T_B)(\rho_{AB})$. The finite-copy criterion makes the counterexample sought by the general problem precise:

$$\exists\, \rho_{AB} : \quad \rho_{AB}^{T_B} \not\geq 0, \qquad \forall\, n \geq 1 \;\; \forall\, |\psi_n\rangle \text{ with } \mathrm{SR}_{A^n:B^n}(|\psi_n\rangle) \leq 2, \quad \langle\psi_n|(\rho_{AB}^{T_B})^{\otimes n}|\psi_n\rangle \geq 0. \tag{1}$$

Here $\mathrm{SR}_{A^n:B^n}$ is Schmidt rank across the indicated cut. A state satisfying Eq. (1) would be NPT bound entangled; equivalently, the question is whether every NPT state instead has a negative expectation on some Schmidt-rank-at-most-two vector at some finite copy number.

The nested low-rank frontier asks whether the counterexample in Eq. (1) can additionally satisfy

$$\mathrm{rank}(\rho_{AB}) = 5. \tag{2}$$

Thus Eq. (2) asks for the lowest rank not already excluded by known one-copy results.

Status

Unsolved

Source

Krüger and Werner explicitly list NPT bound entanglement as an open problem, while Chen and Đoković isolate rank five as the next frontier after proving all rank-at-most-four NPT states distillable [KW05], [CD16].

Progress

- Horodecki, Horodecki, and Horodecki established the finite-copy Schmidt-rank criterion used in Eq. (1) and proved that states with positive partial transpose are undistillable. Their theorem does not prove the converse for NPT states [HHH98].
- Chen and Đoković proved the sharp low-rank obstruction

$$\mathrm{rank}(\rho_{AB}) \leq 4, \quad \rho_{AB}^{T_B} \not\geq 0 \quad \Longrightarrow \quad \rho_{AB} \text{ is 1-distillable}. \tag{3}$$

  Equation (3) makes Eq. (2) the first possible rank for a one-copy-undistillable NPT state [CD16].
- The same work constructed a parameterized family ρ_ϵ of rank-five two-qutrit NPT states and proved the fixed-copy statement, but not the all-copy statement,

$$\underbrace{\forall\, n \geq 1 \; \exists\, \delta_n > 0 \; \forall\, \epsilon \in (0, \delta_n] : \quad \rho_\epsilon \text{ is } n\text{-undistillable}}_{\text{proved}} \quad \not\Longrightarrow \quad \underbrace{\exists\, \epsilon_* > 0 \; \forall\, n \geq 1 : \quad \rho_{\epsilon_*} \text{ is } n\text{-undistillable}}_{\text{needed}}. \tag{4}$$

  The authors conjectured the missing right-hand statement in Eq. (4) [CD16].
- For a different one-parameter family of symmetric rank-five two-qutrit states, Lei, Song, Chen, and Liu proved 1-undistillability on the remaining NPT interval $[(24\sqrt{2}-33)/7, (33-12\sqrt{6})/25)$. For two copies, they proved that any Schmidt-rank-at-most-two negative-expectation vector must have a component outside a specified 17-dimensional subspace. This narrows the search but proves neither two-copy nor all-copy undistillability [LSC+26].
- A principal unrestricted candidate family is formed by the $d \times d$ Werner states

$$\rho_\alpha = \frac{I + \alpha F}{d^2 + \alpha d}, \qquad F|x\rangle|y\rangle = |y\rangle|x\rangle, \qquad -\frac{1}{2} \leq \alpha < -\frac{1}{d}, \quad d \geq 3. \tag{5}$$

  The parameter window in Eq. (5) is exactly the NPT, one-copy-undistillable window in this convention [DSS+00].

- Costa Rico reformulated finite-copy Werner distillability through partial-trace inequalities and obtained new finite-copy bounds [CR25]. Two separate July 2026 preprints subsequently proved that ρ_α is two-copy distillable exactly when $\alpha < -1/2$, in every local dimension. Hence the full candidate window in Eq. (5) is two-copy undistillable, but these results do not control all tensor powers [FHPV26], [BGH26].

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Comment

The unrestricted question is posed explicitly in the source collection [KW05]. The rank-five formulation is a nested but stronger route to a counterexample: a rank-five example would settle the general problem, whereas excluding rank five would leave higher ranks open. The decisive quantifier is the existence of one fixed state that is n-undistillable for every n; separately choosing a state or parameter for each n, as on the left of Eq. (4), is insufficient.

Tag

NPT states; Bound entanglement; Entanglement distillation; Low-rank quantum states; Partial transpose criterion; Local operations and classical communication

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11 Complete facet descriptions for Bell polytopes

Problem. Determine complete facet descriptions for local-behavior polytopes beyond the presently solved Bell scenarios, either for a specified unresolved finite scenario or for a nontrivial infinite family with additional structure. For positive integers N, M, and K, let $\mathbf{x} \in \{1, \ldots, M\}^N$ denote the measurement settings and $\mathbf{a} \in \{1, \ldots, K\}^N$ the outcomes. The relevant local polytope is

$$\mathcal{L}_{N,M,K} := \operatorname{conv}\left\{ p_{\mathbf{f}} : p_{\mathbf{f}}(\mathbf{a} \mid \mathbf{x}) = \prod_{i=1}^{N} \mathbf{1}\{a_i = f_i(x_i)\}, \quad f_i : \{1, \ldots, M\} \to \{1, \ldots, K\} \right\}. \tag{1}$$

Here $\mathbf{1}\{\cdot\}$ is the indicator function. The task is to characterize all facet-defining inequalities of the polytope in Eq. (1) in a chosen unresolved regime, modulo permutations of parties, settings, and outcomes and modulo liftings obtained by adjoining redundant settings or outcomes. A solution must carry a proof of completeness, not merely generate a large collection of facets.

Status

Unsolved

Source

Krüger and Werner explicitly pose the search for completeness-certified facet descriptions of Bell polytopes in tractable finite scenarios and structured families [KW05].

Progress

- Fine proved that positivity, normalization, and the Clauser–Horne–Shimony–Holt inequalities completely describe the $(N, M, K) = (2, 2, 2)$ local scenario. This settles only the smallest nontrivial case [Fin82].
- Werner and Wolf classified all full-correlation facets for arbitrary N when every party has two dichotomic observables. Their theorem concerns a correlator projection of Eq. (1), not the full conditional-probability polytope [WW01].
- Staufenbiel's polyhedral-sampling method found more than 1.29×10^8 inequivalent facet classes in one unresolved scenario, but the method intentionally sacrifices completeness. Thus the computation enlarges the known catalog without supplying the required certificate [Sta26].

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Comment

The source problem explicitly replaces an implausible classification uniform in all (N, M, K) by completeness-certified finite cases and structured families [KW05]. The unresolved gap is a new theorem of completeness for one such nontrivial regime; heuristic completeness or sampling without a certificate is insufficient.

Tag

Bell nonlocality; Convex geometry; Combinatorics; Quantum foundations

ID

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12 Relative entropy of entanglement for two qubits

Problem. Find a closed formula for the relative entropy of entanglement of every two-qubit density operator ρ, including an explicit closest separable state. With $\mathrm{Sep}(\mathbb{C}^2 : \mathbb{C}^2)$ denoting the two-qubit separable states, the quantity is

$$E_R(\rho) := \min_{\sigma \in \mathrm{Sep}(\mathbb{C}^2:\mathbb{C}^2)} D(\rho\|\sigma), \qquad D(\rho\|\sigma) := \begin{cases} \mathrm{Tr}[\rho(\log\rho - \log\sigma)], & \mathrm{supp}\,\rho \subseteq \mathrm{supp}\,\sigma, \\ +\infty, & \text{otherwise.} \end{cases} \tag{1}$$

In the finite branch of Eq. (1), the trace is evaluated on $\mathrm{supp}\,\rho$, with $0\log 0 := 0$. The formula sought must determine at least one minimizing state σ_ρ for every ρ.

Status

Unsolved

Source

The general two-qubit formula is listed by Krüger and Werner, and Miranowicz and Ishizaka explicitly distinguish this unresolved forward problem from their solved inverse construction [KW05], [MI08].

Progress

- For two qubits, separability is equivalent to positivity under partial transpose. Thus Eq. (1) is a convex optimization over the positive-partial-transpose set, but this equivalence does not produce a closed optimizer [HHH96].
- Miranowicz and Ishizaka solved the inverse problem: from a boundary separable state σ, they parameterized entangled states for which σ is closest. Their construction yields formulas for special families but does not invert to a closed map $\rho \mapsto \sigma_\rho$ for an arbitrary input [MI08].
- Friedland and Gour extended the inverse optimality characterization to general dimensions and proved uniqueness of the closest separable state for full-rank entangled inputs. These structural results still do not evaluate Eq. (1) in closed form for every two-qubit state [FG11].

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Comment

The source collection and the full discussion of the inverse construction explicitly identify the forward formula as open [KW05], [MI08]. The unresolved step is a closed determination of σ_ρ from arbitrary input data ρ; special-state formulas and an inverse parameterization do not supply it.

Tag

Entanglement measures; Qubit systems; Convex optimization; Quantum separability

ID

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13 Long-range vacuum CHSH violation

Problem. Does the vacuum of a massive free scalar Bose field violate the CHSH inequality between bounded localization regions at arbitrarily large spacelike separation? Fix a bounded region O in Minkowski spacetime, let O_L be a congruent spacelike translate at separation L, and write $\mathcal{A}(O)$ and $\mathcal{A}(O_L)$ for the commuting local von Neumann algebras. For the vacuum state ω_0, define

$$\beta(L) := \frac{1}{2} \sup_{A,A',B,B'} \left|\omega_0\big(A(B+B') + A'(B-B')\big)\right|, \tag{1}$$

where $A, A' \in \mathcal{A}(O)$ and $B, B' \in \mathcal{A}(O_L)$ range over self-adjoint contractions. With the normalization in Eq. (1), the local hidden-variable bound is 1. The open question is whether

$$\{L > 0 : \beta(L) > 1\} \quad \text{is unbounded.} \tag{2}$$

Equation (2) asks for direct, unfiltered two-setting violation by the vacuum restrictions to the two local algebras.

Status

Unsolved

Source

Werner and Wolf explicitly ask whether vacuum Bell violation persists at arbitrarily large separation; the same question is recorded in the open problem collection of Krüger and Werner [WW01], [KW05].

Progress

- Summers and Werner proved vacuum Bell violation for suitable spacelike separated observables, including maximal violation in important short-range configurations. This does not control the fixed bounded-region geometry as L grows [SW85].
- In a massive theory, clustering makes the excess $\beta(L) - 1$ decay with separation. Werner and Wolf explicitly noted that decay does not determine whether the excess remains positive at every finite distance or becomes exactly zero beyond a threshold [WW01].
- Recent work closes part of the gap between algebraic existence arguments and explicit CHSH observables. Guimaraes, Roditi, and Sorella constructed bounded Hermitian field observables giving vacuum CHSH violation for a massive scalar field in $1+1$ dimensions, but for tangent causal diamonds [GRS25]. Azevedo et al. obtained violations from unitary-deformed sign observables in complementary wedge algebras [AGG+26]. Tangent diamonds have zero gap and wedges are unbounded, so neither construction controls $\beta(L)$ for congruent bounded regions as $L \to \infty$.
- Verch and Werner proved long-range nonclassicality through failure of the analogue of positive partial transpose and through distillability under broad algebraic hypotheses [VW05]. Reznik, Retzker, and Silman obtained Bell violations with localized probes coupled to the vacuum [RRS05]. Filtering, multi-copy distillation, or replacing the field algebras by detector systems is strictly weaker than establishing Eq. (2) for the unfiltered vacuum CHSH value.

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Comment

The source problem asks precisely for the unbounded-distance alternative in Eq. (2) [KW05]. Persistent entanglement, distillability, generic Bell correlation, and detector-assisted violations do not decide the specified single-copy CHSH supremum.

Tag

Algebraic quantum field theory; Bell nonlocality; Bosonic systems; Quantum foundations

ID

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14 Tough error models

Problem. Determine $c(e, n)$ and construct tough error models in the sense of [KW05]. Let $H = \mathbb{C}^n$, and let an e-dimensional error model be a complex linear subspace $E \subseteq \mathrm{End}(H)$ with $1 \le e \le n^2$. A linear subspace $C \subseteq H$ corrects E exactly when, for every $A, B \in E$, there is a scalar $\lambda(A, B) \in \mathbb{C}$ such that

$$P_C A^\dagger B P_C = \lambda(A, B) P_C, \tag{1}$$

where P_C is the orthogonal projector onto C. Define the guaranteed code dimension by

$$c(e, n) := \min_{\substack{E \subseteq \mathrm{End}(H) \\ \dim E = e}} \max \left\{ \dim C : \begin{array}{l} C \subseteq H \text{ is a linear subspace, and} \\ \forall A, B \in E \ \exists \lambda(A, B) \in \mathbb{C} : \ P_C A^\dagger B P_C = \lambda(A, B) P_C \end{array} \right\}. \tag{2}$$

Determine $c(e, n)$ in Eq. (2), exactly or with asymptotically matching bounds, and exhibit explicit tough error models whose largest correcting code is close to this value.

Status

Unsolved

Source

Krüger and Werner explicitly introduce the extremal function $c(e, n)$ and ask for tough error models attaining its scale [KW05].

Progress

- The compression condition in Eq. (1) is the Knill–Laflamme criterion for perfect correction of the linear error space E [KLV00].
- The Knill–Laflamme–Viola existence argument and the projection construction recorded in the source give the ceiling-sensitive bounds
$$\left\lceil \frac{\lceil n/e^2 \rceil}{e^2 + 1} \right\rceil \le c(e, n) \le \left\lceil \frac{n}{e} \right\rceil. \tag{3}$$
The upper bound uses an error space spanned by nearly equal-rank orthogonal projections. The parameter gap is polynomial in e [KLV00], [KW05].
- The one-error case can be settled exactly. For $E = \mathrm{span}\{A\}$, the condition is a scalar compression of the Hermitian matrix $A^\dagger A$. The Hermitian higher-rank numerical-range formula therefore gives
$$c(1, n) = \left\lceil \frac{n}{2} \right\rceil. \tag{4}$$
A positive matrix with simple spectrum makes the bound sharp [CKZ06]. At the other extreme, the upper construction in Eq. (3), together with the trivial one-dimensional code, gives $c(e, n) = 1$ for $e \ge n$.
- Weaver's quantum Turán framework interprets scalar compressions as quantum anticliques [Wea19]. Allen and Kornell's improved low-dimensional anticlique theorem applies to $\mathcal{V}_E := \mathrm{span}(\{I\} \cup \{A^\dagger B : A, B \in E\})$. Since $\dim \mathcal{V}_E \le 5$ when $e = 2$, it gives $c(2, n) \ge \lceil n/9 \rceil$, improving the historical lower bound of order $n/20$ [AK25]. This remains far from the upper bound $\lceil n/2 \rceil$; general operator-system estimates also do not exploit all of the product structure in $\mathcal{V}_E$.

References

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Comment

The source calls an error model E "tough" when the largest code correcting E has dimension close to $c(e, n)$ [KW05]. The unresolved task is to close the bounds in Eq. (3) for $2 \le e < n$ and construct such models at the correct scale. Results for structured Pauli noise or arbitrary operator systems do not determine Eq. (2).

Tag

Quantum error correction; Quantum coding theory; Operator algebras; Matrix analysis

ID

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15 Absolute separability from spectra

Problem. Characterize the spectra of bipartite states that remain separable under every global unitary, and decide whether absolute separability equals absolute positivity under partial transpose in higher local dimensions. Let $2 \le m \le n$ and let $\lambda = (\lambda_1, \ldots, \lambda_{mn})$ be a decreasing probability vector. Define the absolutely separable spectra by

$$\mathrm{ASEP}_{m,n} := \left\{ \lambda : U \operatorname{diag}(\lambda) U^\dagger \text{ is separable on } \mathbb{C}^m \otimes \mathbb{C}^n \text{ for every } U \in \mathrm{U}(mn) \right\}. \tag{1}$$

The relaxation by positivity under partial transpose is

$$\mathrm{APPT}_{m,n} := \left\{ \lambda : \left(U \operatorname{diag}(\lambda) U^\dagger \right)^{T_B} \succeq 0 \text{ for every } U \in \mathrm{U}(mn) \right\}. \tag{2}$$

The task is to characterize the set in Eq. (1) and, for $3 \le m \le n$, decide whether it equals the set in Eq. (2).

Status

Unsolved

Source

Krüger and Werner pose the spectral characterization of absolute separability, and Ahiable, Kothakonda, and Winter explicitly retain the higher-dimensional characterization and ASEP–APPT equality as open [KW05], [AKW26].

Progress

- For two qubits, Verstraete, Audenaert, and De Moor obtained the exact condition

$$\lambda \in \mathrm{ASEP}_{2,2} \quad \Longleftrightarrow \quad \lambda_1 - \lambda_3 - 2\sqrt{\lambda_2 \lambda_4} \le 0. \tag{3}$$

  Equation (3) settles only the smallest bipartite dimension [VAD01].
- Hildebrand gave necessary-and-sufficient linear-matrix-inequality conditions for membership in Eq. (2) in arbitrary finite dimensions. This characterizes the relaxation, not absolute separability [Hil07].
- Johnston proved $\mathrm{ASEP}_{2,n} = \mathrm{APPT}_{2,n}$ for every n, so all cases with a qubit factor are excluded from the remaining problem [Joh13].
- Abellanet-Vidal et al. derived sufficient spectral criteria for absolute separability in arbitrary dimensions by inverting positive maps and taking convex hulls of the resulting regions. These provide computable inner approximations to $\mathrm{ASEP}_{m,n}$, but are not necessary conditions [AMR+25].
- Ahiable, Kothakonda, and Winter established new geometric properties of both spectral sets. In particular, $\mathrm{APPT}_{m,n}$ is a spectrahedron whose faces are all exposed, whereas $\mathrm{ASEP}_{m,n}$ is semialgebraic. They explicitly retain the higher-dimensional characterization and equality as open [AKW26].
- Tran subsequently derived an explicit dimension-dependent upper bound on the purity of every absolutely-PPT state. This is a quantitative outer constraint on $\mathrm{APPT}_{m,n}$, not a spectral characterization, and hence does not settle its equality with $\mathrm{ASEP}_{m,n}$ [Tra26].

References

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Comment

The spectral characterization was posed in the source collection [KW05]. The remaining gap begins when both local dimensions are at least 3: the complete boundary of Eq. (1) is unknown, as is its possible equality with Eq. (2).

Tag

Quantum separability; Partial transpose criterion; Convex geometry; Qudit systems

ID

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16 Collective cost of tensor-power state preparation

Problem. Determine the asymptotic weighted circuit cost of preparing tensor powers of a known pure state, and characterize when collective preparation is cheaper per copy than independent preparation. On an arbitrary qubit register, allow Pauli-product rotations $e^{i\phi P}$ with $\phi \in \mathbb{R}$ and $P \in \{I, X, Y, Z\}^{\otimes q}$, assigning each such gate the cost $|\phi|$. For a known m-qubit state $|\psi\rangle$, let $U(\boldsymbol{\phi}, \boldsymbol{P}) := \prod_{j=1}^{r} e^{i\phi_j P_j}$ for a finite sequence of angles and Pauli products. Define the phase-independent exact n-copy cost by

$$C_n(\psi) := \inf \left\{ \sum_{j=1}^{r} |\phi_j| : \begin{array}{l} r \in \mathbb{N}_0,\ \phi_j \in \mathbb{R},\ P_j \in \{I, X, Y, Z\}^{\otimes mn}\ (1 \le j \le r), \\ U(\boldsymbol{\phi}, \boldsymbol{P})|0^{mn}\rangle\langle 0^{mn}|U(\boldsymbol{\phi}, \boldsymbol{P})^\dagger = (|\psi\rangle\langle\psi|)^{\otimes n} \end{array} \right\}. \tag{1}$$

Determine the growth of Eq. (1) with n and characterize the states for which the regularized cost

$$C_\infty(\psi) := \lim_{n\to\infty} \frac{C_n(\psi)}{n} = \inf_{n\ge 1} \frac{C_n(\psi)}{n} \tag{2}$$

satisfies $C_\infty(\psi) < C_1(\psi)$. For an approximation tolerance $\varepsilon \ge 0$, also determine the scaling of

$$C_n^\varepsilon(\psi) := \inf_U \left\{ \mathrm{cost}(U) : \frac{1}{2} \left\| U|0^{mn}\rangle\langle 0^{mn}|U^\dagger - (|\psi\rangle\langle\psi|)^{\otimes n} \right\|_1 \le \varepsilon \right\}, \tag{3}$$

where U ranges over all finite circuits of Pauli-product rotations on the mn-qubit register and $\mathrm{cost}(U)$ is the corresponding sum of absolute rotation angles, as in Eq. (1). The question for Eq. (3) includes fixed and vanishing error sequences.

Status

Unsolved

Source

The exact and approximate tensor-power preparation questions are posed in the open-problem collection of Krüger and Werner; Scarani et al. independently identify collective product-state preparation complexity as an open direction [KW05], [SIG+05].

Progress

- Independent preparation gives $C_n(\psi) \le nC_1(\psi)$. Concatenation gives $C_{n+k}(\psi) \le C_n(\psi) + C_k(\psi)$, so Fekete's lemma establishes the equality in Eq. (2). These elementary bounds do not decide whether the inequality can be strict.
- Plesch and Brukner gave nearly optimal worst-case state-synthesis circuits on a fixed register. Their discrete local-gate count neither uses the weighted Pauli-string metric in Eq. (1) nor exploits a tensor-power promise [PB11].
- Mora and Briegel related precision-dependent quantum-state algorithmic complexity to entanglement for several state families. Their framework does not determine the direct-product scaling in Eq. (3) [MB05].

References

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Comment

The exact cost model and its approximate variant are posed in the source collection [KW05]; a cloning review also identifies product-preparation complexity as an open direction [SIG+05]. The state is known, so the problem is not universal cloning. The unresolved quantity is the collective saving in Eqs. (2) and (3) for this specific continuous gate metric.

Tag

Quantum state preparation; Quantum circuit complexity; Qubit systems

ID

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17 SIC-POVM existence in every dimension

Problem. Does a symmetric informationally complete positive-operator-valued measure exist in every finite dimension $d \geq 2$? Equivalently, determine whether for every such d there are d^2 unit vectors $|\psi_1\rangle, \ldots, |\psi_{d^2}\rangle \in \mathbb{C}^d$ satisfying

$$\sum_{j=1}^{d^2} |\psi_j\rangle\langle\psi_j| = dI_d, \qquad |\langle\psi_j|\psi_k\rangle|^2 = \frac{1}{d+1} \quad \text{for all } j \neq k. \tag{1}$$

When Eq. (1) holds, the effects $\Pi_j = d^{-1}|\psi_j\rangle\langle\psi_j|$ form the desired SIC-POVM. No covariance or additional symmetry is required.

Status

Unsolved

Source

Renes, Blume-Kohout, Scott, and Caves explicitly conjecture that SIC-POVMs exist in every finite dimension [RBS+04].

Progress

- Renes, Blume-Kohout, Scott, and Caves explicitly conjectured existence in every dimension and found Weyl–Heisenberg-covariant numerical solutions through $d = 45$. Numerical solutions do not prove the universal quantifier in Eq. (1) [RBS+04].
- Appleby, Bengtsson, Flammia, and Goyeneche reported numerical solutions in every dimension through $d = 181$ and in many larger dimensions. The computations provide extensive evidence but no all-dimensional construction [ABFG19].
- Appleby, Flammia, and Kopp constructed SICs in every dimension $d > 3$ conditional on two unproved number-theoretic conjectures. The conditional hypotheses prevent this result from establishing Eq. (1) unconditionally [AFK25].
- A claimed unconditional proof posted in 2026 was withdrawn after the author stated that the proof was incorrect, so it does not change the status of the problem [Jok26].

References

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Comment

The remaining gap is unconditional existence for every integer $d \geq 2$ in Eq. (1). Numerical evidence, covariance-restricted constructions, and results conditional on number-theoretic conjectures do not settle the unrestricted existence question.

Tag

Symmetric informationally complete measurements; Measurement theory; Quantum state tomography; Qudit systems

ID

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18 Weyl–Heisenberg-covariant SICs in every dimension

Problem. Does every finite dimension admit a symmetric informationally complete measurement that is a single Weyl–Heisenberg orbit? For an integer $d \geq 2$, let $\omega_d = e^{2\pi i/d}$ and define shift, phase, and displacement operators on the computational basis by

$$X_d|j\rangle = |j+1 \pmod{d}\rangle, \qquad Z_d|j\rangle = \omega_d^j|j\rangle, \qquad D_{p,q} = X_d^p Z_d^q, \quad (p,q) \in \mathbb{Z}_d^2. \tag{1}$$

Equation (1) fixes a phase convention that does not affect the orbit of rank-one projectors. The question is whether, for every $d \geq 2$, there is a unit vector $|\phi\rangle \in \mathbb{C}^d$ satisfying

$$\left|\langle\phi|D_{p,q}|\phi\rangle\right|^2 = \frac{1}{d+1} \qquad \text{for every } (p,q) \in \mathbb{Z}_d^2 \setminus \{(0,0)\}. \tag{2}$$

If Eq. (2) holds, the d^2 projectors in the Weyl–Heisenberg orbit of $|\phi\rangle$ form a SIC.

Status

Unsolved

Source

Renes, Blume-Kohout, Scott, and Caves explicitly conjecture the existence of a Weyl–Heisenberg-covariant SIC in every finite dimension [RBS+04].

Progress

- Renes, Blume-Kohout, Scott, and Caves stated Eq. (2) in every dimension as Conjecture 1 and found numerical solutions through $d = 45$ [RBS+04].
- Appleby, Bengtsson, Flammia, and Goyeneche reported numerical Weyl–Heisenberg SICs in every dimension through $d = 181$ and in many larger dimensions. These computations establish individual finite cases, not the all-dimensional assertion [ABFG19].
- Appleby, Flammia, and Kopp give a construction for all $d > 3$ conditional on the order-one abelian Stark conjecture and a special-value identity for the Shintani–Faddeev modular cocycle. The hypotheses remain unproved, so the construction is not unconditional [AFK25].

References

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Comment

The unresolved step is an unconditional construction, or existence proof, for Eq. (2) in every d. Problem 19 imposes the additional requirement that the fiducial be Zauner symmetric.

Tag

Symmetric informationally complete measurements; Measurement theory; Qudit systems; Representation theory

ID

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19 Zauner-symmetric Weyl–Heisenberg SIC fiducials

Problem. Does every finite dimension admit a Weyl–Heisenberg SIC fiducial that is an eigenvector of a Zauner Clifford unitary? For $d \geq 2$, define the displacement operators $D_{\mathbf{p}} := D_{p,q} := X_d^p Z_d^q$ for $\mathbf{p} = (p,q)^\intercal \in \mathbb{Z}_d^2$, using

$$X_d|j\rangle = |j+1 \pmod{d}\rangle, \qquad Z_d|j\rangle = e^{2\pi i j/d}|j\rangle, \qquad \mathbf{p} = (p,q)^\intercal \in \mathbb{Z}_d^2. \tag{1}$$

Equation (1) fixes the Weyl–Heisenberg orbit up to irrelevant phases. Let U_Z be a Clifford unitary whose action on displacement operators is

$$U_Z D_{\mathbf{p}} U_Z^\dagger \doteq D_{F_Z \mathbf{p}}, \qquad F_Z = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}, \tag{2}$$

where indices are reduced modulo d and $\doteq$ denotes equality up to phase. Equation (2) fixes the distinguished order-three Clifford symmetry. The question is whether, for every $d \geq 2$, there are a unit vector $|\phi\rangle$ and a phase $e^{i\theta}$ such that

$$U_Z|\phi\rangle = e^{i\theta}|\phi\rangle, \qquad \left|\langle\phi|D_{\mathbf{p}}|\phi\rangle\right|^2 = \frac{1}{d+1} \quad \text{for every } \mathbf{p} \in \mathbb{Z}_d^2 \setminus \{\mathbf{0}\}. \tag{3}$$

Equation (3) simultaneously imposes Zauner symmetry and the SIC overlap equations.

Status

Unsolved

Source

Appleby states the all-dimensional existence of a Zauner-symmetric Weyl–Heisenberg SIC fiducial as Conjecture B [App05].

Progress

- Appleby stated Eq. (3) as Conjecture B and verified it for the numerical fiducials then known, through $d = 45$ [App05].
- Larger exact and numerical data sets continue to exhibit the Zauner symmetry, including Weyl–Heisenberg SIC solutions in every dimension through $d = 181$. Finite computations do not prove the universal quantifier in Eq. (3) [ABFG19].
- Appleby, Flammia, and Kopp construct Zauner-symmetric SICs for all $d > 3$ under two number-theoretic conjectures. Those unproved assumptions leave the unconditional problem open [AFK25].

References

[App05] D. M. Appleby, "SIC-POVMs and the Extended Clifford Group," *Journal of Mathematical Physics* **46**, 052107 (2005). doi:10.1063/1.1896384; arXiv:quant-ph/0412001.

[ABFG19] M. Appleby, I. Bengtsson, S. Flammia, and D. Goyeneche, "Tight Frames, Hadamard Matrices and Zauner's Conjecture," *Journal of Physics A: Mathematical and Theoretical* **52**, 295301 (2019). doi:10.1088/1751-8121/ab25ad; arXiv:1903.06721.

[AFK25] M. Appleby, S. T. Flammia, and G. S. Kopp, "A Constructive Approach to Zauner's Conjecture via the Stark Conjectures," arXiv:2501.03970 (2025). arXiv:2501.03970.

Comment

The open problem is the unconditional all-dimensional existence of a fiducial satisfying both conditions in Eq. (3). It is strictly stronger than Weyl–Heisenberg-covariant SIC existence in Problem 18.

Tag

Symmetric informationally complete measurements; Measurement theory; Qudit systems; Representation theory

ID

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20 Secret key from every entangled state

Problem. Does every finite-dimensional entangled bipartite state have positive asymptotic distillable secret key? For a state ρ_{AB}, with an adversary holding a purification, define its distillable-key rate by

$$K_D(\rho_{AB}) := \sup \left\{ R \geq 0 : \begin{array}{l} \text{there are LOPC protocols } \Lambda_n \text{ and private states } \gamma_{K_n}^{(n)}, \text{ with } K_n = 2^{\lfloor nR \rfloor}, \text{ such that} \\ \lim\limits_{n\to\infty} \left\| \Lambda_n(\rho_{AB}^{\otimes n}) - \gamma_{K_n}^{(n)} \right\|_1 = 0 \end{array} \right\}, \tag{1}$$

where each target $\gamma_{K_n}^{(n)}$ may have its own shield system and has key registers of dimension K_n. With the operational convention in Eq. (1), the question is whether the implication

$$\rho_{AB} \text{ entangled} \quad \Longrightarrow \quad K_D(\rho_{AB}) > 0 \tag{2}$$

holds for every finite-dimensional ρ_{AB}.

Status

Unsolved

Source

Horodecki, Sikorski, Das, and Wilde explicitly identify the question whether every entangled state has positive distillable key [HSDW26].

Progress

- Horodecki, Horodecki, Horodecki, and Oppenheim constructed bound-entangled states with positive distillable key. Thus zero distillable entanglement does not provide a counterexample to Eq. (2) [HHHO05].
- Horodecki, Sikorski, Das, and Wilde explicitly state that the existence of an entangled state with zero distillable key remains open. Their 2026 result therefore confirms the unresolved scope of Eq. (2) close to the present cutoff [HSDW26].

References

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Comment

The unresolved alternative is either to prove Eq. (2) or to construct an entangled state with $K_D = 0$. The question concerns the asymptotic key rate, not one-shot key extraction or distillable entanglement.

Tag

Quantum cryptography; Bound entanglement; Entanglement theory; Local operations and classical communication

ID

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21 Lockability of two-way distillable entanglement

Problem. Can discarding one local qubit reduce two-way distillable entanglement by an arbitrarily large amount? Let $D_{\leftrightarrow}(A:B)_\rho$ denote the asymptotic singlet-distillation rate of ρ_{AB} under local operations and unrestricted two-way classical communication. The question is whether there are finite-dimensional states $\rho^{(r)}_{A_r a:B_r}$, with $\dim a = 2$, such that

$$D_{\leftrightarrow}(A_r a : B_r)_{\rho^{(r)}} - D_{\leftrightarrow}(A_r : B_r)_{\mathrm{Tr}_a\, \rho^{(r)}} \xrightarrow[r\to\infty]{} \infty. \tag{1}$$

Equation (1) requires an unbounded loss while the discarded subsystem has fixed dimension two.

Status

Unsolved

Source

Horodecki et al. prove lockability of one-way distillable entanglement and explicitly leave the two-way quantity unresolved [HHHO05].

Progress

- Horodecki, Horodecki, Horodecki, and Oppenheim constructed families in which discarding one qubit causes an unbounded loss of entanglement of formation, entanglement cost, logarithmic negativity, or one-way distillable entanglement. These results do not establish Eq. (1), which allows unrestricted two-way communication [HHHO05].
- The same work proved that discarding one qubit decreases the relative entropy of entanglement by at most two ebits, but explicitly left lockability of unrestricted distillable entanglement open. No corresponding universal bound is known for $D_{\leftrightarrow}$ [HHHO05].

References

[HHHO05] K. Horodecki, M. Horodecki, P. Horodecki, and J. Oppenheim, "Locking Entanglement with a Single Qubit," *Physical Review Letters* **94**, 200501 (2005). doi:10.1103/PhysRevLett.94.200501; arXiv:quant-ph/0404096.

Comment

The remaining gap is to construct a family satisfying Eq. (1) or prove a dimension-independent upper bound on the loss of $D_{\leftrightarrow}$. The one-way analogue is already known to be lockable.

Tag

Entanglement distillation; Entanglement measures; Local operations and classical communication; Qubit systems

ID

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22 Honest-party lockability of distillable key

Problem. Can loss of one qubit held by an honest party reduce the two-way distillable secret key by an arbitrarily large amount? Let $K_D(A:B)_\rho$ denote the asymptotic secret-key rate obtainable from ρ_{AB} by local operations and public two-way classical communication, with an adversary holding a purification. The question is whether there are finite-dimensional states $\rho^{(r)}_{A_r a:B_r}$, with $\dim a = 2$, such that

$$K_D(A_r a : B_r)_{\rho^{(r)}} - K_D(A_r : B_r)_{\mathrm{Tr}_a\, \rho^{(r)}} \xrightarrow[r\to\infty]{} \infty. \tag{1}$$

Equation (1) is the AB-locking question: the discarded qubit belongs to Alice rather than being transferred to the adversary.

Status

Unsolved

Source

Christandl et al. explicitly ask whether distillable key is lockable when the discarded subsystem is held by an honest party [CEH+07].

Progress

- Christandl and collaborators proved that transferring a bounded subsystem to the adversary cannot cause an unbounded loss of distillable key. They explicitly left the distinct honest-party, or AB-, locking problem open, even for classical tripartite states [CEH+07].
- Horodecki and collaborators established non-lockability bounds for restricted families of private states. They identify the arbitrary-state version of Eq. (1) as unresolved, including a specified product-system special case [HSK+21].

References

[CEH+07] M. Christandl, A. Ekert, M. Horodecki, P. Horodecki, J. Oppenheim, and R. Renner, "Unifying Classical and Quantum Key Distillation," in *Theory of Cryptography Conference 2007*, Lecture Notes in Computer Science **4392**, 456–478 (Springer, 2007). doi:10.1007/978-3-540-70936-7_25; arXiv:quant-ph/0608199.

[HSK+21] K. Horodecki, M. Studziński, R. P. Kostecki, O. Sakarya, and D. Yang, "Upper Bounds on the Leakage of Private Data and an Operational Approach to Markovianity," *Physical Review A* **104**, 052422 (2021). doi:10.1103/PhysRevA.104.052422; arXiv:2107.10737.

Comment

The unresolved scope is AB-lockability of K_D for arbitrary states, as formalized in Eq. (1). It excludes both the non-lockable adversary-side operation and the restricted private-state families covered by existing bounds.

Tag

Quantum cryptography; Entanglement measures; Local operations and classical communication; Resource conversion; Qubit systems

ID

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23 Quantum violations of bipartite Bell facets

Problem. Must every nontrivial facet Bell inequality in a finite bipartite scenario have a quantum violation? Let $\mathcal{L}$ be the local polytope, and let $\mathcal{Q}$ consist of behaviors realizable as

$$p(a, b \mid x, y) = \mathrm{Tr}\Big[\rho_{AB}\left(M_x^a \otimes N_y^b\right)\Big], \tag{1}$$

where ρ_{AB} is a finite-dimensional state and $\{M_x^a\}_a$ and $\{N_y^b\}_b$ are local POVMs. Equation (1) defines the quantum set used below.

For a linear functional $F(p) = \sum_{a,b,x,y} c_{abxy} p(a, b \mid x, y)$, suppose $F(p) \le \beta_{\mathrm{L}}$ supports a facet of $\mathcal{L}$ and is not a positivity facet within the normalization and no-signalling affine hull. Is it necessarily true that

$$\sup_{p \in \mathcal{Q}} F(p) > \beta_{\mathrm{L}}? \tag{2}$$

Equation (2) asks whether the bipartite local and quantum sets can share a nontrivial facet-supporting hyperplane.

Status

Unsolved

Source

Ramanathan explicitly asks whether every nontrivial bipartite facet Bell inequality, rather than only every such almost-quantum facet, admits a quantum violation [Ram21].

Progress

- Escolà-Farràs, Calsamiglia, and Winter proved Eq. (2) for every tight bipartite correlation inequality, including every facet inequality of a two-player XOR game. Their theorem does not cover general Bell behaviors with arbitrary outcome probabilities [ECW20].
- Ramanathan proved that every nontrivial bipartite facet is violated by the almost-quantum set and that the quantum statement holds for the correlation-polytope projection. Because the almost-quantum set strictly contains the quantum set, this does not establish Eq. (2) in general [Ram21].

References

[ECW20] L. Escolà-Farràs, J. Calsamiglia, and A. Winter, "All Tight Correlation Bell Inequalities Have Quantum Violations," *Physical Review Research* **2**, 012044(R) (2020). doi:10.1103/PhysRevResearch.2.012044; arXiv:1908.06669.

[Ram21] R. Ramanathan, "Violation of All Two-Party Facet Bell Inequalities by Almost-Quantum Correlations," *Physical Review Research* **3**, 033100 (2021). doi:10.1103/PhysRevResearch.3.033100; arXiv:2004.07673.

Comment

The missing step is to replace the almost-quantum violation by a quantum behavior for every nontrivial bipartite facet. A solution may instead exhibit a facet for which $\sup_{p \in \mathcal{Q}} F(p) = \beta_{\mathrm{L}}$, rather than the strict inequality in Eq. (2).

Tag

Bell nonlocality; Convex geometry; Quantum foundations

ID

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24 Optimal CGLMP measurements for a maximally entangled state

Problem. For every $d \geq 3$, do the standard Fourier–phase measurements optimize the CGLMP violation of the maximally entangled state among all projective d-outcome measurements? Fix the state

$$|\Phi_d\rangle = \frac{1}{\sqrt{d}} \sum_{j=0}^{d-1} |j, j\rangle. \tag{1}$$

Equation (1) is held fixed; the shared state is not part of the optimization.
For outcomes in $\mathbb{Z}_d$, let $[t]_d \in \{0, \ldots, d-1\}$ be the residue of t and define the CGLMP functional by

$$\begin{aligned} B_d = \mathbb{E}([A_0 - B_0]_d) + \mathbb{E}([B_0 - A_1]_d) \\ + \mathbb{E}([A_1 - B_1]_d) + \mathbb{E}([B_1 - A_0 - 1]_d). \end{aligned} \tag{2}$$

Local behaviors satisfy $B_d \geq d - 1$ under the convention in Eq. (2). The candidate measurements have bases

$$\begin{aligned} |a; x\rangle &= \frac{1}{\sqrt{d}} \sum_{j=0}^{d-1} \exp\left(\frac{2\pi i}{d} j(a + \alpha_x)\right) |j\rangle, \\ |b; y\rangle &= \frac{1}{\sqrt{d}} \sum_{j=0}^{d-1} \exp\left(\frac{2\pi i}{d} j(-b + \beta_y)\right) |j\rangle, \end{aligned} \tag{3}$$

with $\alpha_0 = 0$, $\alpha_1 = -1/2$, $\beta_0 = 1/4$, and $\beta_1 = 3/4$. The question is whether the bases in Eq. (3) minimize Eq. (2) on Eq. (1), up to symmetries of the state and functional.

Status

Unsolved

Source

Zohren and Gill formulate the standard Fourier-phase measurements as the candidate global optimizers of the CGLMP value for a fixed maximally entangled state [ZG08].

Progress

- Collins, Gisin, Linden, Massar, and Popescu introduced the high-dimensional Bell family and the Fourier–phase measurement construction underlying Eq. (3) [CGLMP02].
- Zohren and Gill explicitly identify optimality of Eq. (3) for the fixed state in Eq. (1) as a longstanding conjecture and provide numerical support. They do not prove it for arbitrary d [ZG08].
- For $d = 3$, Lang, Vértesi, and Navascués obtained a semidefinite upper bound on the optimum over maximally entangled states that differs from the candidate value by about 10^{-10}. This numerical certification is not an exact proof and does not address all dimensions [LVN14].

References

[CGLMP02] D. Collins, N. Gisin, N. Linden, S. Massar, and S. Popescu, "Bell Inequalities for Arbitrarily High-Dimensional Systems," *Physical Review Letters* **88**, 040404 (2002). doi:10.1103/PhysRevLett.88.040404; arXiv:quant-ph/0106024.

[ZG08] S. Zohren and R. D. Gill, "Maximal Violation of the Collins-Gisin-Linden-Massar-Popescu Inequality for Infinite Dimensional States," *Physical Review Letters* **100**, 120406 (2008). doi:10.1103/PhysRevLett.100.120406; arXiv:quant-ph/0612020.

[LVN14] B. Lang, T. Vértesi, and M. Navascués, "Closed Sets of Correlations: Answers from the Zoo," *Journal of Physics A: Mathematical and Theoretical* **47**, 424029 (2014). doi:10.1088/1751-8113/47/42/424029; arXiv:1402.2850.

Comment

The unresolved statement is global measurement optimality for every finite d with the state fixed by Eq. (1). It does not ask for the unrestricted CGLMP optimum over shared states, which is a different problem.

Tag

Bell nonlocality; Measurement theory; Qudit systems; Convex optimization

ID

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25 Statistical strength of CGLMP measurements

Problem. For every $d \geq 3$, do the standard CGLMP Fourier–phase measurements maximize the relative-entropy statistical strength against local realism among all projective d-outcome measurements on the fixed state $|\Phi_d\rangle = d^{-1/2} \sum_{j=0}^{d-1} |j,j\rangle$, when the setting distribution is also optimized? For a behavior $p(a,b \mid x,y)$, a distribution $\mu(x,y)$ on the four setting pairs, and the local polytope $\mathcal{L}$, define

$$S(p;\mu) := \inf_{\ell \in \mathcal{L}} \sum_{a,b,x,y} \mu(x,y) p(a,b \mid x,y) \log_2 \frac{p(a,b \mid x,y)}{\ell(a,b \mid x,y)}, \tag{1}$$

Set $S^\star(p) := \sup_{\mu \in \Delta(\{0,1\}^2)} S(p;\mu)$; by Eq. (1), this is the optimized asymptotic evidence rate against the best local model. The candidate bases are

$$\begin{aligned} |a;x\rangle &= \frac{1}{\sqrt{d}} \sum_{j=0}^{d-1} \exp\left(\frac{2\pi i}{d} j(a+\alpha_x)\right) |j\rangle, \\ |b;y\rangle &= \frac{1}{\sqrt{d}} \sum_{j=0}^{d-1} \exp\left(\frac{2\pi i}{d} j(-b+\beta_y)\right) |j\rangle, \end{aligned} \tag{2}$$

where $\alpha_0 = 0$, $\alpha_1 = -1/2$, $\beta_0 = 1/4$, and $\beta_1 = 3/4$. Let p_{CGLMP} denote the behavior produced on $|\Phi_d\rangle$ by the bases in Eq. (2), and write p_M for the behavior produced by any other measurement choice M on $|\Phi_d\rangle$. The conjectured optimality of Eq. (2) is

$$S^\star(p_{\mathrm{CGLMP}}) = \sup_M S^\star(p_M), \tag{3}$$

where the supremum is over two projective d-outcome measurements per party. Equation (3) is the question to be resolved.

Status

Unsolved

Source

Gill explicitly proposes the global statistical-strength optimality of the CGLMP measurement construction for a fixed maximally entangled state [Gil07].

Progress

- Van Dam, Gill, and Grünwald established the operational interpretation of Eq. (1) as the asymptotic evidence rate of a Bell experiment and formulated the joint optimization problem [DGG05].
- Acín, Gill, and Gisin numerically found the CGLMP measurement bases for their relative-entropy searches at $d = 3$ and $d = 4$. Their globally best state was not maximally entangled, so this does not resolve the fixed-state equality in Eq. (3) [AGG05].
- Gill reports that numerical searches with the maximally entangled state fixed found only the CGLMP measurements for both Euclidean and relative-entropy criteria, but treats optimality as conjectural; the search does not prove Eq. (3) for arbitrary d [Gil07].

References

[DGG05] W. van Dam, R. D. Gill, and P. D. Grünwald, "The Statistical Strength of Nonlocality Proofs," *IEEE Transactions on Information Theory* **51**, 2812–2835 (2005). doi:10.1109/TIT.2005.851738; arXiv:quant-ph/0307125.

[AGG05] A. Acín, R. Gill, and N. Gisin, "Optimal Bell Tests Do Not Require Maximally Entangled States," *Physical Review Letters* **95**, 210402 (2005). doi:10.1103/PhysRevLett.95.210402; arXiv:quant-ph/0506225.

[Gil07] R. D. Gill, "Better Bell Inequalities (Passion at a Distance)," in *Asymptotics: Particles, Processes and Inverse Problems*, IMS Lecture Notes–Monograph Series **55**, 135–148 (2007). doi:10.1214/074921707000000328; arXiv:math/0610115.

Comment

The remaining gap is a global proof or counterexample to Eq. (3) for arbitrary d. The shared state is fixed; this differs from joint state–measurement optimization and from maximizing the linear CGLMP violation in Problem 24.

Tag

Bell nonlocality; Convex optimization; Measurement theory; Qudit systems; Quantum foundations

ID

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26 Gaussian entanglement of formation beyond bisymmetry

Problem. Does Gaussian entanglement of formation equal unrestricted entanglement of formation for every non-bisymmetric multimode Gaussian state? Let ρ_{AB} be a Gaussian state of $n_A + n_B \geq 3$ bosonic modes, with covariance matrix V. Its entanglement of formation is

$$E_F(\rho_{AB}) := \inf_{\rho_{AB}=\sum_j p_j|\psi_j\rangle\langle\psi_j|} \sum_j p_j S(\mathrm{Tr}_B|\psi_j\rangle\langle\psi_j|)\,, \tag{1}$$

where the infimum is over all pure-state ensembles. The Gaussian restriction of Eq. (1) is equivalently

$$E_F^G(\rho_{AB}) := \inf_{\substack{V_p \preceq V \\ V_p \text{ pure Gaussian}}} E(V_p), \tag{2}$$

where $E(V_p)$ is the entropy of either reduced state of the pure Gaussian state with covariance matrix V_p. A covariance matrix is bisymmetric when it is invariant under arbitrary permutations of Alice's modes and, independently, of Bob's modes. The question is whether $E_F(\rho_{AB}) = E_F^G(\rho_{AB})$ outside this bisymmetric family; the two quantities are fixed by Eqs. (1) and (2).

Status

Unsolved

Source

Adesso proves Gaussian optimality for all two-mode and bisymmetric multimode states and explicitly leaves the general nonsymmetric multimode case open [Ade26].

Progress

- Wolf, Giedke, Krüger, Werner, and Cirac introduced the Gaussian convex-roof restriction and derived the covariance-matrix optimization in Eq. (2). Their analysis does not establish equality with the unrestricted roof in Eq. (1) [WGKWC04].
- Adesso proved $E_F = E_F^G$ for every two-mode Gaussian state and for every bisymmetric multimode Gaussian state. The same work explicitly leaves the generic nonsymmetric multimode case open [Ade26].

References

[WGKWC04] M. M. Wolf, G. Giedke, O. Krüger, R. F. Werner, and J. I. Cirac, "Gaussian Entanglement of Formation," *Physical Review A* **69**, 052320 (2004). doi:10.1103/PhysRevA.69.052320; arXiv:quant-ph/0306177.

[Ade26] G. Adesso, "Optimality of Gaussian Entanglement of Formation," arXiv:2608.01909v2 (2026). doi:10.48550/arXiv.2608.01909; arXiv:2608.01909v2.

Comment

As of August 31, 2026, equality of the two convex roofs is proved for all two-mode states and for bisymmetric multimode states, but neither a proof nor a counterexample is known for a general non-bisymmetric state with at least three modes.

Tag

Gaussian quantum information; Continuous-variable systems; Entanglement measures; Bosonic systems

ID

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27 One-bit simulation of partially entangled qubits

Problem. Can shared randomness and one classical bit exactly simulate every pair of local projective measurements on every pure entangled two-qubit state? In Schmidt form the state is

$$|\psi_p\rangle = \sqrt{p}\,|00\rangle + \sqrt{1-p}\,|11\rangle, \qquad \frac{1}{2} < p < 1. \tag{1}$$

For arbitrary Bloch vectors $\mathbf{x}, \mathbf{y} \in S^2$ and outcomes $a, b \in \{0,1\}$, the target distribution associated with Eq. (1) is

$$\begin{aligned} P_p(a, b \mid \mathbf{x}, \mathbf{y}) &= \operatorname{Tr}\big[|\psi_p\rangle\langle\psi_p|\big(M_{a|\mathbf{x}} \otimes M_{b|\mathbf{y}}\big)\big]\,, \\ M_{a|\mathbf{x}} &= \frac{I + (-1)^a \mathbf{x}\cdot\boldsymbol{\sigma}}{2}, \qquad M_{b|\mathbf{y}} = \frac{I + (-1)^b \mathbf{y}\cdot\boldsymbol{\sigma}}{2}. \end{aligned} \tag{2}$$

The protocol may use unlimited shared randomness; after receiving $\mathbf{x}$, Alice sends Bob one bit, and their local outputs must reproduce Eq. (2) for every pair of measurement directions. If no such universal protocol exists, find a finite-setting linear inequality satisfied by all one-bit protocols and violated by one of these target distributions.

Status

Unsolved

Source

Gisin explicitly asks whether one classical bit simulates every partially entangled two-qubit state; Renner and Quintino restate the surviving question after narrowing the parameter range [Gis09], [RQ23].

Progress

- Gisin posed the separating-inequality question for one-bit-assisted correlations [Gis09]. At the maximally entangled endpoint $p = 1/2$, Toner and Bacon gave an exact one-bit protocol for all projective measurements [TB03]; this does not cover the nonmaximally entangled interval in Eq. (1).
- Renner and Quintino constructed a one-trit protocol for every $1/2 \le p \le 1$. For $1/2 < p < 1$, their protocol uses only one bit whenever
$$\frac{2p(1-p)}{2p-1}\log\left(\frac{p}{1-p}\right) + 2(1-p) \le 1, \tag{3}$$
which holds approximately for $0.835 \le p < 1$; the product-state endpoint $p = 1$ requires no communication. Thus Eq. (3) leaves the interval $1/2 < p < 0.835$ unresolved [RQ23].
- Numerical and semianalytical models support one-bit simulation beyond the proved range, but do not supply an exact protocol for all directions [SLYS23]. Conversely, finite-scenario inequalities for one-bit-assisted correlations can be enumerated, but no known inequality separates the remaining targets in Eq. (2) [BT03].

References

[Gis09] N. Gisin, "Bell Inequalities: Many Questions, a Few Answers," in W. C. Myrvold and J. Christian (eds.), *Quantum Reality, Relativistic Causality, and Closing the Epistemic Circle*, The Western Ontario Series in Philosophy of Science **73**, 125–138 (Springer, 2009). doi:10.1007/978-1-4020-9107-0_9; arXiv:quant-ph/0702021.

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[SLYS23] P. Sidajaya, A. D. Lim, B. Yu, and V. Scarani, "Neural Network Approach to the Simulation of Entangled States with One Bit of Communication," *Quantum* **7**, 1150 (2023). doi:10.22331/q-2023-10-24-1150; arXiv:2305.19935.

[BT03] D. Bacon and B. F. Toner, "Bell Inequalities with Auxiliary Communication," *Physical Review Letters* **90**, 157904 (2003). doi:10.1103/PhysRevLett.90.157904; arXiv:quant-ph/0208057.

Comment

The unresolved parameter range is approximately $1/2 < p < 0.835$. It is not known whether one bit always suffices there or whether a finite-setting inequality can prove that it does not.

Tag

Bell nonlocality; Quantum communication complexity; Qubit systems

ID

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28 Universal simulation with two PR boxes

Problem. Can shared randomness and at most two Popescu–Rohrlich boxes exactly simulate every pair of local projective measurements on every two-qubit state, without communication? A Popescu–Rohrlich box is the binary nonsignalling behavior

$$P_{\mathrm{PR}}(u, v \mid s, t) = \begin{cases} \frac{1}{2}, & u \oplus v = st, \\ 0, & u \oplus v \neq st, \end{cases} \qquad s, t, u, v \in \{0, 1\}. \tag{1}$$

The parties may use arbitrary local, possibly adaptive wirings of two independent copies of Eq. (1). By convexity and Schmidt decomposition, it suffices to solve the simulation for all pure states

$$|\psi_\theta\rangle = \cos\theta\,|00\rangle + \sin\theta\,|11\rangle, \qquad 0 < \theta \leq \frac{\pi}{4}, \tag{2}$$

and all local projective measurements on the state in Eq. (2). A complementary finite-scenario objective is to characterize the convex set generated by two-box wirings, for example through its facet inequalities.

Status

Unsolved

Source

Gisin explicitly asks whether two Popescu–Rohrlich boxes suffice to simulate all projective-measurement correlations of arbitrary two-qubit pure states [Gis09].

Progress

- Gisin explicitly posed both the universal two-box simulation problem and the finite-scenario inequality problem [Gis09].
- One copy of Eq. (1), together with shared randomness, exactly simulates arbitrary projective measurements on a maximally entangled qubit pair [CGMP05]. This covers only the endpoint $\theta = \pi/4$ in Eq. (2).
- Finite-setting inequalities valid for all one-PR-box strategies are violated by correlations of nonmaximally entangled two-qubit states. Hence one PR box is not a universal resource, but these inequalities do not decide whether two boxes suffice [BGS05], [BSG06].
- Photonic experiments have violated one-box bounds and thereby certified a two-box lower bound for selected measurements. This does not construct a two-box simulation valid for every state and measurement [CLBGK15].

References

[Gis09] N. Gisin, "Bell Inequalities: Many Questions, a Few Answers," in W. C. Myrvold and J. Christian (eds.), *Quantum Reality, Relativistic Causality, and Closing the Epistemic Circle*, The Western Ontario Series in Philosophy of Science **73**, 125–138 (Springer, 2009). doi:10.1007/978-1-4020-9107-0_9; arXiv:quant-ph/0702021.

[CGMP05] N. J. Cerf, N. Gisin, S. Massar, and S. Popescu, "Simulating Maximal Quantum Entanglement without Communication," *Physical Review Letters* **94**, 220403 (2005). doi:10.1103/PhysRevLett.94.220403; arXiv:quant-ph/0410027.

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[CLBGK15] B. G. Christensen, Y.-C. Liang, N. Brunner, N. Gisin, and P. G. Kwiat, "Exploring the Limits of Quantum Nonlocality with Entangled Photons," *Physical Review X* **5**, 041052 (2015). doi:10.1103/PhysRevX.5.041052; arXiv:1506.01649.

Comment

Neither a universal exact two-box protocol nor a counterexample requiring more than two boxes is known. In finite input–output scenarios, a complete description of the convex set generated by two-box wirings is likewise open.

Tag

Bell nonlocality; Resource conversion; Qubit systems

ID

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29 Finite-alphabet nonsignalling simulation of entangled qubits

Problem. For every partially entangled two-qubit pure state, does some fixed finite-alphabet nonsignalling resource give an exact noncommunicating simulation of all local projective measurements? Fix

$$|\psi_\theta\rangle = \cos\theta\,|00\rangle + \sin\theta\,|11\rangle, \qquad 0 < \theta < \frac{\pi}{4}. \tag{1}$$

The target correlations for Bloch directions $\mathbf{x}, \mathbf{y} \in S^2$ are

$$P_\theta(a, b \mid \mathbf{x}, \mathbf{y}) = \mathrm{Tr}\left[|\psi_\theta\rangle\langle\psi_\theta|\left(M_{a|\mathbf{x}} \otimes M_{b|\mathbf{y}}\right)\right], \qquad a, b \in \{0, 1\}, \tag{2}$$

where $M_{a|\mathbf{x}} = (I + (-1)^a \mathbf{x} \cdot \boldsymbol{\sigma})/2$ and similarly for Bob. For each θ in Eq. (1), one may choose a nonsignalling box $R(u, v \mid s, t)$ with finite input and output alphabets and a finite number of copies of it. The box, the number of copies, and the local wiring may depend on θ but not on $\mathbf{x}$ or $\mathbf{y}$, and the wiring may use unlimited shared randomness but no communication. It must reproduce Eq. (2) exactly.

Status

Unsolved

Source

Gisin explicitly poses finite-alphabet nonsignalling simulation, while Brunner et al. isolate the same finiteness gap after giving a continuous-input construction [Gis09], [BGPS08].

Progress

- Gisin posed the existence of a finite-input, finite-output nonsignalling resource for this task [Gis09].
- At the excluded maximally entangled endpoint $\theta = \pi/4$, one finite-alphabet PR box and shared randomness reproduce all local projective measurements exactly [CGMP05].
- Brunner, Gisin, Popescu, and Scarani constructed an exact simulation for every state in Eq. (1), but their protocol uses a continuous-input millionaire box that compares real parameters. They explicitly leave replacement by finite-input resources open [BGPS08].
- Some finite measurement sets obtained from partially entangled states cannot be simulated with a single PR box. This rules out one particular resource budget, not all finite-alphabet resources or all finite numbers of uses [BGS05].

References

[Gis09] N. Gisin, "Bell Inequalities: Many Questions, a Few Answers," in W. C. Myrvold and J. Christian (eds.), *Quantum Reality, Relativistic Causality, and Closing the Epistemic Circle*, The Western Ontario Series in Philosophy of Science **73**, 125–138 (Springer, 2009). doi:10.1007/978-1-4020-9107-0_9; arXiv:quant-ph/0702021.

[CGMP05] N. J. Cerf, N. Gisin, S. Massar, and S. Popescu, "Simulating Maximal Quantum Entanglement without Communication," *Physical Review Letters* **94**, 220403 (2005). doi:10.1103/PhysRevLett.94.220403; arXiv:quant-ph/0410027.

[BGPS08] N. Brunner, N. Gisin, S. Popescu, and V. Scarani, "Simulation of Partial Entanglement with Nonsignaling Resources," *Physical Review A* **78**, 052111 (2008). doi:10.1103/PhysRevA.78.052111; arXiv:0803.2359.

[BGS05] N. Brunner, N. Gisin, and V. Scarani, "Entanglement and Non-locality Are Different Resources," *New Journal of Physics* **7**, 88 (2005). doi:10.1088/1367-2630/7/1/088; arXiv:quant-ph/0412109.

Comment

Continuous-input nonsignalling simulation is known, whereas exact simulation by any finite-alphabet box used finitely many times is not. The finiteness requirement applies to both input and output alphabets and to the number of resource uses.

Tag

Bell nonlocality; Resource conversion; Qubit systems

ID

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30 Secret key from every Bell-nonlocal behavior

Problem. Does every finite-alphabet Bell-nonlocal behavior have a strictly positive asymptotic secret-key rate against arbitrary individual nonsignalling attacks? Let $P(a, b \mid x, y)$ be bipartite and nonsignalling, and suppose that it has no local decomposition of the form

$$P(a, b \mid x, y) = \int_{\Lambda} \mu(d\lambda)\, P_A(a \mid x, \lambda) P_B(b \mid y, \lambda). \tag{1}$$

Thus Eq. (1) fails for every probability measure μ and local response functions P_A, P_B. Alice and Bob receive independent copies of P; on each copy, an adversary may hold an arbitrary nonsignalling extension. The honest parties may choose their inputs, process all outputs locally, and communicate publicly. The question asks whether some such protocol always extracts secret key at a nonzero asymptotic rate.

Status

Unsolved

Source

Gisin explicitly asks whether every Bell-nonlocal behavior can yield secret key against nonsignalling individual attacks [Gis09].

Progress

- Gisin posed the implication from Bell nonlocality to positive secret key specifically for nonsignalling adversaries restricted to individual attacks [Gis09].
- Bell nonlocality is necessary for secrecy in this adversarial model, and explicit nonlocal families admit positive-key protocols. These results do not prove sufficiency for every behavior that violates Eq. (1) [AGM06].
- Some nonlocal correlations obtained from noisy two-qubit Werner states have zero key rate for the broad class of standard device-independent protocols that announce measurements before classical postprocessing. This is not an impossibility theorem for all protocols or for the full individual nonsignalling-adversary model [FBJLA21].
- Bipartite bound information exists for finite classical sources: there are distributions with zero key rate and positive formation cost. Those sources have no Bell inputs and hence do not settle the implication for Bell-nonlocal behaviors [PGR26].

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Comment

The unresolved implication concerns unrestricted protocols at the behavior level under arbitrary individual nonsignalling extensions. Impossibility for a standard protocol class and zero-key classical sources are strictly narrower statements.

Tag

Bell nonlocality; Device-independent quantum information; Quantum key distribution; Quantum cryptography

ID

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31 Semialgebraicity of closed quantum correlations

Problem. Is the closure of the finite-dimensional tensor-product correlation set semialgebraic in every fixed bipartite Bell scenario? Fix positive integers n_A, n_B, m_A, m_B. Inputs satisfy $1 \le x \le n_A$ and $1 \le y \le n_B$, while outputs satisfy $1 \le a \le m_A$ and $1 \le b \le m_B$. The set $\mathcal{C}_q(n_A, n_B, m_A, m_B)$ consists of behaviors

$$p(a, b \mid x, y) = \operatorname{Tr}\big[\rho\big(E_a^x \otimes F_b^y\big)\big] \tag{1}$$

obtained from arbitrary finite-dimensional local Hilbert spaces, a density operator ρ, and local POVMs satisfying

$$E_a^x \succeq 0, \qquad F_b^y \succeq 0, \qquad \sum_{a=1}^{m_A} E_a^x = I, \qquad \sum_{b=1}^{m_B} F_b^y = I. \tag{2}$$

Equation (2) fixes the measurement class in Eq. (1). Define the closed correlation set by

$$\mathcal{C}_{qa}(n_A, n_B, m_A, m_B) := \overline{\mathcal{C}_q(n_A, n_B, m_A, m_B)}. \tag{3}$$

For every fixed tuple, is the set in Eq. (3) a finite Boolean combination of polynomial equalities and inequalities with real coefficients? The coefficients may be arbitrary real numbers, and the description need not be one conjunction of weak inequalities.

Status

Unsolved

Source

Tsirelson explicitly asks whether the quantum-correlation set in each fixed finite Bell scenario is semialgebraic [Tsi93].

Progress

- In the scenario with two binary measurements per party, Jordan-type dimension reduction and finite convexification give a semialgebraic parametrization [Tsi93], [Mas05]. This does not extend to every fixed input–output tuple.
- The zero-marginal correlator slice in the smallest bipartite scenario has an explicit semialgebraic description but is not basic semialgebraic. Consequently, asking for one finite conjunction of polynomial inequalities would already be too strong [LMSWZ23].
- The finite-dimensional set $\mathcal{C}_q$ defined by Eqs. (1)–(2) is not closed in sufficiently large fixed scenarios. This necessitates the closure in Eq. (3) [Slo19].
- For sufficiently large fixed input and output sizes, membership in $\mathcal{C}_{qa}$ is undecidable even for behaviors with algebraic coordinates. This excludes effective polynomial descriptions with computable algebraic coefficients, but it does not exclude a non-effective semialgebraic presentation using arbitrary real coefficients [FMS25].

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Comment

Undecidable membership rules out effective descriptions of the usual kind, but semialgebraicity with unrestricted real coefficients is a set-theoretic existence question and remains unresolved for general fixed scenarios.

Tag

Bell nonlocality; Convex geometry; Operator algebras

ID

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32 The PPT-squared conjecture

Problem. Must the composition of any two compatible PPT completely positive maps be entanglement breaking? Let $\Phi : M_{d_1}(\mathbb{C}) \to M_{d_2}(\mathbb{C})$ and $\Psi : M_{d_2}(\mathbb{C}) \to M_{d_3}(\mathbb{C})$ be completely positive. The Choi operator of Φ is

$$J(\Phi) := \sum_{i,j=1}^{d_1} |i\rangle\langle j| \otimes \Phi(|i\rangle\langle j|), \tag{1}$$

and $J(\Psi)$ is defined analogously. In the convention of Eq. (1), a map Φ is PPT when

$$(T \otimes \mathrm{id})\big(J(\Phi)\big) \succeq 0, \tag{2}$$

and it is entanglement breaking when $J(\Phi)$ is separable; the same definitions apply to Ψ. Under condition Eq. (2) for both maps, is $J(\Psi \circ \Phi)$ necessarily separable? Equivalently, does postselection on any joint measurement outcome on the middle systems of two PPT bipartite states always leave a separable state on the two outer systems?

Status

Unsolved

Source

Christandl, Müller–Hermes, and Wolf formulate the PPT-squared statement as an explicit conjecture about compositions of PPT maps [CMW19].

Progress

- The conjecture was recorded in the Banff workshop report on operator structures in quantum information [BIRS12].
- Christandl, Müller–Hermes, and Wolf formulated the two-map statement above, proved its equivalence to the self-composition version, and established it in equal dimension $d = 2$, for Gaussian channels, and for several further special classes. Their eventual entanglement-breaking results for repeated composition do not imply the two-composition claim [CMW19].
- The equal-dimension conjecture holds for $d = 3$ [CYT19]. Consequently, equal dimension $d \geq 4$ is the first unresolved regime.
- The conjecture holds for diagonal-unitary-covariant maps, a class containing Choi-type, depolarizing, dephasing, and amplitude-damping examples [SN22]. This covariance restriction does not cover arbitrary PPT maps.
- Other repeated-composition theorems prove eventual entanglement breaking for broad PPT families, but not after exactly two maps [KMP18].
- A 2026 qutrit theorem proves a stronger composition statement for cones larger than the PPT cone and extends it to arbitrary selective measurements. Its dimension-specific hypothesis leaves higher dimensions open [AL26].

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Comment

The two-composition claim is settled in equal dimensions two and three and for several structured families. No proof or counterexample is known for arbitrary compatible finite-dimensional PPT maps, with equal dimension four being the first open square case.

Tag

Entanglement-breaking channels; Choi states; Partial transpose criterion; Quantum channels

ID

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33 POVM steering threshold of higher-dimensional Werner states

Problem. What is the exact steering threshold for arbitrary POVMs on a higher-dimensional Werner state? For $d \geq 3$, let F be the swap operator on $\mathbb{C}^d \otimes \mathbb{C}^d$ and define

$$\rho_f^{(d)} := \frac{(d-f)I + (df-1)F}{d(d^2-1)}, \qquad -1 \leq f \leq 1. \tag{1}$$

If Alice applies a POVM $\{M_{a|x}\}_a$ to the state in Eq. (1), Bob's subnormalized conditional states are

$$\sigma_{a|x} := \mathrm{Tr}_A\left[(M_{a|x} \otimes I)\rho_f^{(d)}\right]. \tag{2}$$

The assemblage in Eq. (2) is unsteerable when it admits a local-hidden-state decomposition

$$\sigma_{a|x} = \int_\Lambda \mu(d\lambda)\, p(a \mid x, \lambda)\tau_\lambda, \tag{3}$$

where μ is a probability measure, $p(a \mid x, \lambda)$ are response functions, and τ_λ are density operators. Determine the critical value $f_{\mathrm{POVM}}(d)$ characterized by

$$\rho_f^{(d)} \text{ is unsteerable from Alice to Bob for every POVM} \quad \Longleftrightarrow \quad f \geq f_{\mathrm{POVM}}(d). \tag{4}$$

In particular, decide whether the threshold in Eq. (4) equals the exact projective-measurement threshold

$$f_{\mathrm{PVM}}(d) = -1 + \frac{1}{d} + \frac{1}{d^2} \tag{5}$$

for every $d \geq 3$. Equation (3) fixes the notion of unsteerability used in both threshold statements.

Status

Unsolved

Source

After resolving the qubit case, Zhang–Chitambar and Renner explicitly leave the arbitrary-POVM steering threshold of higher-dimensional Werner states open [ZC24], [Ren24].

Progress

- The projective-measurement boundary is exactly Eq. (5) [Wer89], [JWD07].
- Explicit local-hidden-state constructions give nontrivial arbitrary-POVM unsteerable intervals in every dimension, but for $d \geq 3$ their bounds do not reach Eq. (5) [NG20].
- For $d = 2$, projective measurements and arbitrary POVMs have the same exact threshold. In the visibility parametrization $\rho_W(r) = r|\Psi^-\rangle\langle\Psi^-| + (1-r)I/4$, the state is unsteerable for every POVM exactly when $r \leq 1/2$ [ZC24], [Ren24]. These qubit constructions have not determined Eq. (4) for $d \geq 3$.
- Higher-dimensional steering inequalities detect steering for selected measurement families, but do not determine the all-POVM boundary [YQ26].

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Comment

The exact arbitrary-POVM boundary is known in dimension two but not for any general $d \geq 3$. The remaining question is whether genuinely nonprojective POVMs lower the unsteerable Werner interval below the projective threshold.

Tag

Quantum steering; Qudit systems; Measurement theory; Entanglement theory

ID

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34 Trace-exponential lower bound for matrix-word averages

Problem. Is the normalized trace average of all words in two positive-definite matrices always bounded below by the corresponding trace exponential? Let $A, B \in M_d(\mathbb{C})$ be positive definite, let $n, m \geq 1$, and let $\mathcal{W}_{n,m}$ be the set of words containing exactly n letters A and m letters B. Define the normalized average by

$$p_{n,m}(A,B) := \frac{1}{\binom{n+m}{n}} \sum_{W \in \mathcal{W}_{n,m}} \operatorname{Tr} W(A,B). \tag{1}$$

The question is whether the quantity in Eq. (1) satisfies

$$p_{n,m}(A,B) \geq \operatorname{Tr}\exp\bigl(n \log A + m \log B\bigr) \tag{2}$$

for every finite d and all $n, m \geq 1$. A positive-semidefinite extension is obtained, whenever the limit exists, by applying Eq. (2) to $A + \varepsilon I$ and $B + \varepsilon I$ and then taking $\varepsilon \downarrow 0$.

Status

Unsolved

Source

Cha and Lee formulate a two-sided refined BMV inequality and disprove only its upper half, leaving the lower trace-exponential comparison stated here open [CL26].

Progress

- Equality holds in Eq. (2) when A and B commute. If $n = 1$ or $m = 1$, cyclicity makes every summand in Eq. (1) equal to $\operatorname{Tr}(A^n B^m)$, and the Golden–Thompson inequality proves Eq. (2) [Gol65], [Tho65]. Thus the first unresolved range has $n, m \geq 2$.
- The proved Bessis–Moussa–Villani coefficient-positivity theorem implies only $p_{n,m}(A,B) \geq 0$, which is weaker than Eq. (2) [LS04], [Sta13].
- The original refinement also proposed the distinct upper comparison $\operatorname{Tr}(A^n B^m) \geq p_{n,m}(A,B)$. Cha and Lee disproved that comparison with 3×3 positive-semidefinite matrices at $n = m = 5$ and obtained an unbounded ratio $p_{5,5}(A,B)/\operatorname{Tr}(A^5 B^5)$ [CL26]. Their construction does not disprove Eq. (2); Dinh's subsequent pinching proposal also concerns a replacement for the failed upper comparison [Din26].

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Comment

Cha and Lee state the two-sided refinement explicitly and disprove only its upper half. The lower comparison in Eq. (2) has neither a general proof nor a counterexample.

Tag

Matrix analysis; Combinatorics

ID

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35 Extensible causality and process purification

Problem. Is every finite-dimensional extensibly causal process matrix purifiable? Call an N-laboratory process matrix W extensibly causal if $W \otimes \rho_R$ generates only causal correlations for every ancillary input system R, every joint ancillary state ρ_R, and every choice of local instruments. Call W purifiable if there exist auxiliary global past and future systems, a fixed pure state on the auxiliary past, and a pure process S such that inserting the fixed state into S and discarding the auxiliary future yields W. If EC_N and Pur_N denote the corresponding classes, the question is whether

$$\mathrm{EC}_N \subseteq \mathrm{Pur}_N \tag{1}$$

holds for every N and every choice of finite local dimensions. In particular, does the inclusion in Eq. (1) hold for two laboratories?

Status

Unsolved

Source

Araújo, Feix, Navascués, and Brukner explicitly identify the possible inclusion of extensibly causal processes in the purifiable class as open [AFN+17].

Progress

- Extensible causality rules out activation of noncausal correlations by entangled ancillary inputs. Ordinary causality is insufficient: there are causal processes that become noncausal after such an extension [OG16].
- Necessary algebraic conditions for purifiability exclude several process matrices that violate causal inequalities, but remain inconclusive for the noisy processes that motivated Eq. (1) [AFN+17].
- For two laboratories, every purifiable process is extensibly causal, so $\mathrm{Pur}_2 \subseteq \mathrm{EC}_2$ [YQS+21]. This is the converse of the open implication in Eq. (1); it neither proves the inclusion nor establishes equality.

References

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Comment

Araújo, Feix, Navascués, and Brukner explicitly identify Eq. (1) as an unresolved possibility and give the noisy process built from their Eq. (37) as a concrete test case. The open direction is extensible causality implying purifiability, not the proved two-laboratory converse and not causal separability.

Tag

Process matrices; Indefinite causal order; Quantum causal inference; Superchannels

ID

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36 Constant trace-distance separability testing

Problem. What is the computational complexity of testing bipartite separability with a constant trace-distance promise gap? For local dimension d, define

$$\mathrm{Sep}(d,d) := \mathrm{conv}\Big\{\alpha \otimes \beta : \alpha, \beta \in \mathcal{D}(\mathbb{C}^d)\Big\}. \tag{1}$$

Fix a constant $0 < \varepsilon_0 < 1$. Given a rational description of $\rho \in \mathcal{D}(\mathbb{C}^d \otimes \mathbb{C}^d)$, promised that exactly one of the following alternatives holds,

$$\begin{aligned} \text{YES:} \quad & \rho \in \mathrm{Sep}(d,d), \\ \text{NO:} \quad & \inf_{\sigma \in \mathrm{Sep}(d,d)} \frac{1}{2}\|\rho - \sigma\|_1 \geq \varepsilon_0, \end{aligned} \tag{2}$$

decide which case in Eq. (2) holds. Is there an algorithm polynomial or quasipolynomial in d? More generally, when the gap ε is part of the input, determine the optimal dependence of the complexity on d and ε.

Status

Unsolved

Source

Harrow and Montanaro explicitly pose constant-gap weak membership for separability in trace norm as an open complexity problem [HM13].

Progress

- Harrow and Montanaro formulate the constant trace-distance promise in Eq. (2) explicitly and relate it to estimating acceptance probabilities in $\mathsf{QMA}(2)$ [HM13].
- Weak membership for the set in Eq. (1) is strongly NP-hard when the promised distance is inverse-polynomial in the input dimension [Gha10]. This hardness regime does not classify the fixed constant gap in Eq. (2).
- A symmetric-extension algorithm has running time

$$\exp\big(O\big(\varepsilon^{-2}(\log d)^2\big)\big) \tag{3}$$

  for Euclidean distance or an operational LOCC norm [BCY11]. The bound in Eq. (3) does not hold as stated for trace distance.
- For each fixed constant gap, randomized polynomial time is now known for Euclidean-norm weak membership [Mal26]. Constant trace distance can coexist with Euclidean distance that vanishes with d, so this result does not solve Eq. (2).

References

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Comment

The constant-gap trace-norm task in Eq. (2) is posed explicitly in Section 4.2, item 14 of [HM13]. The norm is essential: the constant-gap Euclidean problem has been solved, but that result does not classify trace-norm weak membership.

Tag

Quantum separability; Quantum complexity theory; Convex optimization; Semidefinite programming; Entanglement theory

ID

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37 Resources for implementing Gibbs-preserving channels

Problem. What are the exact one-shot coherence and work costs of implementing an arbitrary Gibbs-preserving channel by thermal operations? Let finite systems S and S' have Hamiltonians H_S and $H_{S'}$ at inverse temperature β, with Gibbs states

$$\gamma_X := \frac{e^{-\beta H_X}}{\mathrm{Tr}(e^{-\beta H_X})}, \qquad X \in \{S, S'\}. \tag{1}$$

A channel $\Phi : S \to S'$ is Gibbs preserving when it maps the first state in Eq. (1) to the second. For approximation error $\varepsilon \geq 0$, define its quantum-Fisher-information coherence cost by

$$F_c^\varepsilon(\Phi) := \inf_{\substack{(R,H_R),\ \eta_R \in \mathcal{D}(R) \\ \mathcal{T}: S\otimes R \to S' \text{ thermal}}} \left\{ F_Q(\eta_R, H_R) : D_{\mathrm{ch}}(\Phi, \mathcal{T}(\,\cdot \otimes \eta_R)) \leq \varepsilon \right\}, \tag{2}$$

where the infimum ranges over finite resource systems R with Hamiltonian H_R, and $\mathcal{T}$ is a thermal operation implemented with Gibbs ancillas and an energy-conserving unitary; set $\inf \varnothing := +\infty$. The distance D_{ch} is the supremum of purified distance between the two channel outputs over inputs entangled with an arbitrary reference. If $\eta_R = \sum_j \lambda_j |j\rangle\langle j|$, the quantity in Eq. (2) is

$$F_Q(\eta_R, H_R) := 2 \sum_{j,k:\, \lambda_j + \lambda_k > 0} \frac{(\lambda_j - \lambda_k)^2}{\lambda_j + \lambda_k} \left|\langle j|H_R|k\rangle\right|^2. \tag{3}$$

Equation (3) fixes the normalization of the coherence measure used in Eq. (2). Determine Eq. (2), up to matching bounds, for every Gibbs-preserving Φ and every ε. Give the analogous tight deterministic work cost when an ideal battery begins and ends in sharp energy states and the battery's energy loss is charged as work. Which structural features of Φ determine whether the exact costs are finite, and what is their sharp divergence as $\varepsilon \downarrow 0$ when they are infinite at zero error?

Status

Unsolved

Source

Tajima and Takagi explicitly leave a tight channel-by-channel characterization of coherence and work costs for Gibbs-preserving operations open [TT25].

Progress

- Every thermal operation is Gibbs preserving and time-translation covariant, whereas a general Gibbs-preserving channel need not be covariant and may create energy coherence. The operation classes are therefore inequivalent [FOR15].
- Universal optimal work rates are known in the independent and identically distributed limit, and one-shot implementations are known for important time-covariant channels [FBB21]. These results do not determine either cost requested above for every Gibbs-preserving channel.
- Some Gibbs-preserving channels cannot be implemented exactly by a thermal operation aided by any finite amount of coherence. For pairwise reversible channels, lower bounds on Eq. (2) scale as $1/\varepsilon^2$ and are essentially attained by suitable examples [TT25]. This disproves universal finite coherence sufficiency but does not give a tight formula for arbitrary channels.

References

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Comment

The conclusion of [TT25] explicitly leaves a complete tight characterization of coherence cost for all Gibbs-preserving operations and the corresponding work-cost problem. Universal exact implementation with finite coherence is already known to be false; the open task is the channel-by-channel cost characterization.

Tag

Quantum thermodynamics; Thermal operations; Quantum resource theories; Asymmetry and quantum reference frames; Quantum channels; One-shot quantum information

ID

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38 Polynomial shared-resource lower bounds for routing

Problem. Does an explicit total Boolean family require polynomial shared-state cost for bounded-error one-round f-routing? More precisely, do there exist constants $c > 0$ and $\varepsilon > 0$ and a sequence

$$f_n : \{0,1\}^n \times \{0,1\}^n \longrightarrow \{0,1\} \tag{1}$$

such that one uniform deterministic algorithm computes $f_n(x, y)$ from (n, x, y) in time polynomial in n, and every routing protocol for the map in Eq. (1) with worst-case diamond-norm error at most ε has cost at least n^c?

In an f-routing protocol, Alice receives x and an unknown qubit Q, Bob receives y, and they may share an arbitrary state ρ_{LR} before the inputs arrive. After one simultaneous message in each direction, Alice must recover Q when $f(x, y) = 0$, and Bob must recover it when $f(x, y) = 1$. Message sizes and local operations are unrestricted. Measure only the shared state by

$$E_{\dim}(\rho_{LR}) := \log_2 \min\{\operatorname{rank} \rho_L, \operatorname{rank} \rho_R\}\,. \tag{2}$$

The target is a family in Eq. (1) for which every valid protocol satisfies $E_{\dim}(\rho_{LR}) \geq n^c$, with the cost defined in Eq. (2).

Status

Unsolved

Source

Bogner explicitly asks for polynomial, or even superlogarithmic for an explicit total family, shared-resource lower bounds for f-routing [Bog26].

Progress

- Every Boolean function admits an f-routing protocol with worst-case shared-resource cost at most

$$2^{O(\sqrt{n \log n})}. \tag{3}$$

  The construction underlying Eq. (3) also relates routing quantitatively to conditional disclosure of secrets [ABM+24].
- Rank methods give nontrivial lower bounds for explicit functions when one routing case is required to be perfectly correct, but do not establish a polynomial lower bound on Eq. (2) in the two-sided bounded-error model [ACM24].
- For the explicit inner-product function, the best robust result is

$$f_n(x, y) = \bigoplus_{i=1}^{n} x_i y_i, \qquad d \log_2(2d) = \Omega(n), \qquad E_{\dim} \geq \log_2 n - \log_2 \log_2 n - O(1), \tag{4}$$

  where $d = \min\{\operatorname{rank} \rho_L, \operatorname{rank} \rho_R\}$ and the worst-case error is at most 0.09 [Bog26]. Equation (4) is logarithmic rather than polynomial in the cost measure of Eq. (2).

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Comment

Bogner explicitly leaves a polynomial lower bound on Eq. (2) open and notes that even a superlogarithmic bound for an explicit total family is unknown. The quantity $E_{\dim}$ is a rank-based shared-resource cost for arbitrary mixed states, not an entanglement monotone.

Tag

Position-based quantum cryptography; Distributed quantum computing; Quantum communication complexity; Quantum communication; Quantum cryptography

ID

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39 Unconditional classical verification with one quantum prover

Problem. Does every language $L \in \mathsf{BQP}$ admit a single-prover interactive proof with a fully classical verifier, an efficient quantum honest prover, and information-theoretic soundness? Precisely, require a probabilistic classical polynomial-time verifier exchanging only classical messages with one prover; a uniform quantum polynomial-time honest prover; acceptance probability at least $2/3$ for every yes-instance when the honest prover is used; and acceptance probability at most $1/3$ for every no-instance against every prover, including a computationally unbounded one. Equivalently, can one classically verify an arbitrary efficient quantum computation with one efficient quantum prover, no quantum capability for the verifier, and no cryptographic assumption?

Status

Unsolved

Source

Aharonov, Ben-Or, Eban, and Mahadev explicitly ask whether the verifier's remaining quantum register can be eliminated from an unconditional single-prover verification protocol [ABE+17].

Progress

- The inclusions $\mathsf{BQP} \subseteq \mathsf{PSPACE} = \mathsf{IP}$ give classical-verifier interactive proofs for BQP, but do not ensure that the honest prover is implementable in quantum polynomial time [ADH97], [Sha92].
- Unconditional protocols with an honest BQP prover are known when the verifier retains a constant-size quantum register or exchanges a small number of qubits [ABE+17]. They do not meet the fully classical verifier and classical-message requirements.
- A fully classical single-verifier protocol is known under the learning-with-errors assumption [Mah18]. Its soundness is computational against efficient quantum provers, rather than information theoretic against arbitrary provers.
- Fully classical and unconditional verification is possible with multiple noncommunicating entangled provers [RUV13]. This changes the single-prover model required in the problem statement.

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Comment

Section 1.8 of [ABE+17] explicitly asks whether the verifier's remaining quantum register can be eliminated. The unresolved conjunction is one prover, a fully classical verifier, a BQP honest prover, and unconditional soundness; relaxing any one of these requirements leads to known protocols.

Tag

Interactive proof systems; Delegated quantum computation; Verification and validation; Quantum complexity theory

ID

`op_3770ad932d54692f`

40 Existence of an eight-ququart perfect tensor

Problem. Does an absolutely maximally entangled state of eight ququarts exist? More precisely, with $[8] = \{1, \ldots, 8\}$, determine whether there is a unit vector $|\psi\rangle \in (\mathbb{C}^4)^{\otimes 8}$ such that

$$\mathrm{Tr}_{S^c}(|\psi\rangle\langle\psi|) = \frac{I_{4^4}}{4^4} \qquad \text{for every } S \subseteq [8] \text{ with } |S| = 4. \tag{1}$$

Condition (1) is equivalently the existence of a rank-eight perfect tensor of bond dimension 4: every flattening across a $4|4$ partition is proportional to a unitary [PYHP15]. Under the pure-code correspondence, it is also equivalent to a pure quantum MDS code with parameters $[\![8, 0, 5]\!]_4$ [SZZL26].

Status

Unsolved

Source

Shi, Zhang, Zhao, and Li complete the seven-party AME classification and explicitly identify $\mathrm{AME}(8, 4)$ as the first remaining homogeneous existence case [SZZL26].

Progress

- Bernal proved that a minimally supported $\mathrm{AME}(N, 4)$ state exists only for $N \leq 6$. Thus any solution of (1) must have strictly more than 4^4 nonzero coefficients in every local product basis [Ber19].
- Wójcik, Makuta, Bruzda, and Augusiak proved that no canonical $\mathbb{Z}_d$ graph state can be $\mathrm{AME}(N, d)$ when 4 divides N and d is even. This excludes canonical $\mathbb{Z}_4$ graph-state constructions for (1), but their theorem does not exclude stabilizer constructions over $\mathbb{F}_4$ or general non-stabilizer states [WMB+26].
- Shi, Zhang, Zhao, and Li proved that $\mathrm{AME}(7, d)$ exists exactly for $d \geq 3$, completing the homogeneous existence classification through seven parties. They identify $\mathrm{AME}(8, 4)$ as the first remaining open case [SZZL26].

References

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Comment

The unresolved question concerns arbitrary complex states satisfying (1). Existing no-go results cover minimal-support states and canonical $\mathbb{Z}_4$ graph states, but do not rule out non-minimal-support $\mathbb{F}_4$ stabilizer states or non-stabilizer constructions.

Tag

Absolutely maximally entangled states; Multipartite entanglement; Qudit systems; Quantum coding theory; Tensor networks

ID

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41 Real four-quhex absolutely maximally entangled state

Problem. Does there exist an AME(4, 6) state whose coefficients are real in a product basis? Fix $[6] = \{0, \ldots, 5\}$ and ask for a normalized state

$$|\psi\rangle = \sum_{i,j,k,l\in[6]} T_{ijkl} |i\rangle_A |j\rangle_B |k\rangle_C |l\rangle_D, \qquad T_{ijkl} \in \mathbb{R}, \qquad \sum_{i,j,k,l\in[6]} T_{ijkl}^2 = 1. \tag{1}$$

The state in (1) must have maximally mixed reductions on every pair of parties: for each $S \subseteq \{A, B, C, D\}$ with $|S| = 2$,

$$\rho_S := \mathrm{Tr}_{S^c} |\psi\rangle\langle\psi| = \frac{I_{36}}{36}. \tag{2}$$

Equivalently, define the 36×36 flattening U and its reshuffling U^R and second-factor partial transpose U^Γ by

$$U_{(i,j),(k,l)} := 6T_{ijkl}, \qquad (U^R)_{(i,k),(j,l)} := U_{(i,j),(k,l)}, \qquad (U^\Gamma)_{(i,j),(k,l)} := U_{(i,l),(k,j)}. \tag{3}$$

Here the three matrices in (3) are the coefficient flattenings for the bipartitions $AB|CD$, $AC|BD$, and $AD|CB$, up to the displayed ordering of tensor factors. Consequently, (2) is equivalent to the orthogonal 2-unitarity conditions

$$U \in O(36), \qquad U^R \in O(36), \qquad U^\Gamma \in O(36). \tag{4}$$

Thus the problem is also to construct an orthogonal 2-unitary matrix of order 36, or prove that none exists.

Status

Unsolved

Source

Rajchel-Mieldzioć et al. explicitly ask whether a real AME(4, 6) state, equivalently an orthogonal 2-unitary matrix of order 36, exists [RBR+26].

Progress

- Rather, Burchardt, Bruzda, Rajchel-Mieldzioć, Lakshminarayan, and Życzkowski constructed an exact AME(4, 6) state, equivalently a complex 2-unitary matrix of order 36. Its entries use nontrivial twentieth-root phases. Their numerical searches found high-entangling-power orthogonal matrices but no orthogonal 2-unitary, leading them to conjecture nonexistence in $O(36)$; this is numerical evidence, not a proof [RBB+22].
- Cha proved, for the generalized-Pauli stabilizer convention used in that work, that no stabilizer AME(4, 6) state exists. Hence any real solution of (2) must lie outside that stabilizer class, but the theorem does not exclude real nonstabilizer states [Cha26].

References

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Comment

Item T2 of [RBR+26] explicitly retains the real AME(4, 6) problem and its orthogonal 2-unitary formulation. The precise unresolved gap is whether the three simultaneous orthogonality conditions in (4) have any real solution; neither the known complex constructions nor the stabilizer obstruction answers this question.

Tag

Absolutely maximally entangled states; Multipartite entanglement; Qudit systems; Matrix analysis

ID

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42 Minimal-support frontier for absolutely maximally entangled states

Problem. Determine, for every integer $d \geq 2$, the largest number $\mathcal{N}(d)$ of d-level parties that admit an absolutely maximally entangled state of minimal computational-basis support. To make this precise, let $m = \lfloor N/2 \rfloor$ and write a normalized state as $|\psi\rangle = \sum_{\boldsymbol{x} \in [d]^N} c_{\boldsymbol{x}} |\boldsymbol{x}\rangle$, where $[d] = \{0, \ldots, d-1\}$. The state is $\mathrm{AME}(N, d)$ when every subsystem $S \subseteq \{1, \ldots, N\}$ with $|S| \leq m$ has reduced state

$$\mathrm{Tr}_{S^c} |\psi\rangle\langle\psi| = \frac{I_{d^{|S|}}}{d^{|S|}}. \tag{1}$$

Condition (1) implies the computational-basis support bound

$$\left|\mathrm{supp}_{\mathrm{comp}}(\psi)\right| := \left|\{\boldsymbol{x} : c_{\boldsymbol{x}} \neq 0\}\right| \geq d^m. \tag{2}$$

Minimal support means equality in (2), and the frontier to be determined is

$$\mathcal{N}(d) := \max\bigl\{N \geq 2 : \text{a minimal-support } \mathrm{AME}(N, d) \text{ state exists}\bigr\}. \tag{3}$$

Equivalently, for any alphabet $\mathcal{A}$ of size d, the existence event in (3) is characterized by

$$\begin{aligned} &\text{a minimal-support } \mathrm{AME}(N, d) \text{ state exists} \\ \iff \quad &\exists \mathcal{C} \subseteq \mathcal{A}^N : \quad |\mathcal{C}| = d^{\lfloor N/2 \rfloor}, \qquad \min_{\substack{\boldsymbol{x}, \boldsymbol{y} \in \mathcal{C} \\ \boldsymbol{x} \neq \boldsymbol{y}}} d_{\mathrm{H}}(\boldsymbol{x}, \boldsymbol{y}) = \left\lceil \frac{N}{2} \right\rceil + 1. \end{aligned} \tag{4}$$

Thus (4) asks for general, possibly nonlinear, MDS codes rather than only linear codes [GAL+15], [Ber19].

Status

Unsolved

Source

Bernal formulates the minimal-support AME frontier through its equivalent maximum-distance-separable-code problem and obtains only a conditional general determination [Ber19].

Progress

- Goyeneche et al. proved the MDS equivalence (4), while Bernal proved that, for fixed d, the admissible party numbers form the initial interval $2 \leq N \leq \mathcal{N}(d)$ [GAL+15], [Ber19]. This reduces the frontier problem to a maximum-length question for unrestricted d-ary MDS codes, which is not known in general.
- Bernal obtained the exact small-alphabet values and the prime-power lower bound

$$\begin{gathered} \mathcal{N}(2) = 3, \quad \mathcal{N}(3) = 4, \quad \mathcal{N}(4) = 6, \quad \mathcal{N}(5) = 6, \quad \mathcal{N}(6) = 3, \quad \mathcal{N}(7) = 8, \\ \mathcal{N}(d) \geq d + 1 \quad \text{for every prime power } d \geq 3. \end{gathered} \tag{5}$$

  The value $\mathcal{N}(4) = 6 = d + 2$ in (5) comes from the even-field exceptional MDS parameters at combinatorial dimension 3; consequently, equality with $d + 1$ cannot hold uniformly over prime powers [Ber19].
- Let $M(k, d)$ denote the maximum length of any, not necessarily linear, d-ary MDS code of cardinality d^k. Bernal proved the conditional implication

$$\left.\begin{array}{c} d \geq 8 \text{ is a prime power}, \qquad k_0 = \left\lfloor \dfrac{d+2}{2} \right\rfloor, \\ M(k_0, d) = d + 1 \end{array}\right\} \implies \mathcal{N}(d) = d + 1. \tag{6}$$

  The general MDS conjecture supplies the premise in (6). Its possible $d + 2$ exceptions for $d = 2^j$ and $k \in \{3, d-1\}$ do not apply here because $3 < k_0 < d - 1$ for $d \geq 8$; the missing step is the conjecture for general nonlinear codes [Ber19].

References

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Comment

The unresolved task is the exact existence frontier, not the classification of states realizing it. Support is measured in the fixed computational basis used in (2), and the equivalent coding problem allows arbitrary alphabets and nonlinear codes. The linear MDS conjecture alone therefore does not settle (3).

Tag

Absolutely maximally entangled states; Multipartite entanglement; Qudit systems; Quantum coding theory; Combinatorics

ID

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43 Finite nontrivial LU moduli of AME states

Problem. Do there exist integers $N, d \geq 2$ for which the absolutely maximally entangled states of N qudits of local dimension d form finitely many, but more than one, local-unitary equivalence classes? A normalized vector $|\psi\rangle \in (\mathbb{C}^d)^{\otimes N}$ is an AME(N, d) state when every subsystem of at most half the parties is maximally mixed:

$$\mathrm{Tr}_{S^c}(|\psi\rangle\langle\psi|) = \frac{I_{d^{|S|}}}{d^{|S|}} \quad \text{for every } S \subseteq \{1, \ldots, N\} \text{ with } |S| \leq \left\lfloor \frac{N}{2} \right\rfloor . \tag{1}$$

For normalized states satisfying Eq. (1), define local-unitary equivalence by

$$|\psi\rangle \sim_{\mathrm{LU}} |\phi\rangle \quad \Longleftrightarrow \quad |\psi\rangle = (U_1 \otimes \cdots \otimes U_N)|\phi\rangle \quad \text{for some } U_1, \ldots, U_N \in U(d). \tag{2}$$

If $\mathcal{A}_{N,d}$ denotes the set specified by Eq. (1), determine whether the quotient under Eq. (2) can satisfy

$$\mathfrak{M}_{N,d} := \mathcal{A}_{N,d}/\sim_{\mathrm{LU}}, \qquad 1 < |\mathfrak{M}_{N,d}| < \infty. \tag{3}$$

Status

Unsolved

Source

Rajchel-Mieldzioć et al. explicitly ask whether an AME family can have a finite number of local-unitary equivalence classes greater than one [RBR+26].

Progress

- For four parties, all AME$(4, 3)$ states lie in one LU class, whereas AME$(4, d)$ has infinitely many LU classes for every $d \geq 4$ [RRKL23]. The latter theorem includes infinitely many inequivalent AME$(4, 6)$ states, so neither side of this classification realizes Eq. (3).
- Tan completely classifies five-qubit AME states: every such state is LU equivalent to a point of the unique $((5, 2, 3))$ code $\mathcal{C}$, and two points of $\mathcal{C}$ are equivalent exactly when related by its finite binary-tetrahedral transversal group [Tan26]. Because the projective state space of the two-dimensional code is a continuum while this group is finite, $|\mathfrak{M}_{5,2}|$ is infinite, not finite and nontrivial.
- The 2026 AME review records the known one-class and infinite-class regimes and explicitly leaves Eq. (3) as open problem T5 [RBR+26].

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Comment

The remaining task is either to exhibit an AME parameter pair whose full LU quotient is a discrete non-singleton set, or to prove that every nonempty $\mathfrak{M}_{N,d}$ is a singleton or infinite. Classifying only a selected construction family would not determine the full quotient in Eq. (3).

Tag

Absolutely maximally entangled states; Multipartite entanglement; Entanglement theory; Local unitary equivalence; Qudit systems

ID

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44 Maximum number of mutually unbiased bases

Problem. For each integer $d \geq 2$, determine the maximum number $\mu(d)$ of pairwise mutually unbiased orthonormal bases of $\mathbb{C}^d$. Two orthonormal bases $\mathcal{B}_r = \{|e_i^{(r)}\rangle\}_{i=1}^d$ and $\mathcal{B}_s = \{|e_j^{(s)}\rangle\}_{j=1}^d$ are mutually unbiased when

$$\left|\langle e_i^{(r)}|e_j^{(s)}\rangle\right|^2 = \frac{1}{d} \qquad \text{for every } i, j \in \{1, \dots, d\}. \tag{1}$$

Thus the extremal quantity defined by Eq. (1) is

$$\mu(d) := \max\Big\{m : \text{there exist } m \text{ orthonormal bases of } \mathbb{C}^d \text{ that are pairwise mutually unbiased}\Big\}. \tag{2}$$

Determine Eq. (2) in the non-prime-power regime, in particular for $d \in \{6, 10, 12, 14, 15\}$.

Status

Unsolved

Source

Krüger and Werner explicitly pose determination of the maximum number of mutually unbiased bases in arbitrary dimension; McNulty and Weigert retain the composite-dimensional cases as open [KW05], [MW26].

Progress

- The universal dimension bound is $\mu(d) \leq d + 1$, and finite-field constructions attain it whenever d is a prime power. Hence $\mu(d) = d + 1$ throughout the prime-power regime [WF89], [KR04].
- If $d = \prod_r p_r^{a_r}$ is the prime-power factorization, tensor products give $\mu(d) \geq 1 + \min_r p_r^{a_r}$. Moreover, any family of d mutually unbiased bases extends to $d + 1$ bases. Consequently $\mu(6) \in \{3, 4, 5, 7\}$; in particular, six bases cannot be maximal [KR04], [Wei13].
- No fourth mutually unbiased basis in $\mathbb{C}^6$ is known. Numerical searches support $\mu(6) = 3$, while symmetry-reduced semidefinite hierarchies provide convergent certificate frameworks but have not resolved the unrestricted case [BH07], [GP24], [MW26].

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Comment

Determining the exact value in Eq. (2) is stronger than deciding whether a complete family of $d + 1$ bases exists. For example, excluding seven bases in dimension six would not distinguish among the three remaining possibilities $\mu(6) \in \{3, 4, 5\}$. This extremal MUB problem is distinct from the SIC existence and symmetry questions in Problems 17–19.

Tag

Mutually unbiased bases; Measurement theory; Quantum state tomography; Combinatorics; Semidefinite programming

ID

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45 Unclassified existence parameters for homogeneous AME states

Problem. For which of the parameter pairs specified below does an absolutely maximally entangled state exist? A normalized vector $|\psi\rangle \in (\mathbb{C}^d)^{\otimes n}$ is an AME(n,d) state when

$$\mathrm{Tr}_{S^c}(|\psi\rangle\langle\psi|) = \frac{I_{d^{|S|}}}{d^{|S|}} \quad \text{for every } S \subseteq \{1,\dots,n\} \text{ with } |S| \le \left\lfloor \frac{n}{2} \right\rfloor. \tag{1}$$

Determine whether a state satisfying Eq. (1) exists for every pair in

$$\mathcal{U} := \big\{(8,6),(8,10),(9,6),(9,10),(10,6),(10,10),(11,3),(11,6),(11,10),(12,6),(12,10)\big\}. \tag{2}$$

Thus the task is to classify every pair in Eq. (2) by existence or nonexistence, without restricting to stabilizer or minimal-support states.

Status

Unsolved

Source

The unresolved parameter list is drawn from the AME existence table and open tasks compiled by Rajchel-Mieldzioć et al., with cases removed when later constructions settled them [RBR+26].

Progress

- Existence under Eq. (1) is equivalent to a pure one-dimensional quantum error-correcting code with parameters $((n,1,\lfloor n/2\rfloor+1))_d$. Quantum-code bounds and shadow inequalities therefore provide nonexistence tests, but they do not settle the pairs in Eq. (2) [Sco04], [HG20].
- General constructions yield AME$(5,d)$ and AME$(6,d)$ for every $d \ge 2$, while the qubit existence problem is completely classified: an AME$(n,2)$ state exists exactly for $n \in \{2,3,5,6\}$. These results delimit, but do not decide, Eq. (2) [RBR+26].
- Two 2026 advances remove stale benchmark cases: the seven-party classification proves AME$(7,d)$ exists exactly when $d \ge 3$, and an exact Hermitian self-dual MDS code constructs AME$(12,5)$. Neither result decides any pair retained in Eq. (2) [SZZL26], [BB26].

References

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Comment

The set in Eq. (2) is the nonredundant remainder of the audited DeepMind benchmark list: the solved pairs $(7,6)$, $(7,10)$, and $(12,5)$ are excluded, while the unresolved $(8,4)$ case is already Problem 40. Problems 41–43 respectively impose a real-coefficient constraint, ask for minimal support, and classify local-unitary orbits; none duplicates the unrestricted existence predicates collected here.

Tag

Absolutely maximally entangled states; Multipartite entanglement; Qudit systems; Quantum coding theory; Combinatorics

ID

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46 Fixed-error parallel Stein lemma for quantum channels

Problem. Let $\mathcal{N}, \mathcal{M} : \mathcal{L}(A) \to \mathcal{L}(B)$ be quantum channels on finite-dimensional systems. For states ρ and σ, define the relative entropy and the hypothesis-testing divergence by

$$\begin{aligned} D(\rho\|\sigma) &:= \mathrm{Tr}[\rho(\log_2 \rho - \log_2 \sigma)]\,, \\ D_H^{\varepsilon}(\rho\|\sigma) &:= -\log_2 \inf_{\substack{0 \le Q \le I \\ \mathrm{Tr}(Q\rho) \ge 1-\varepsilon}} \mathrm{Tr}(Q\sigma), \end{aligned} \tag{1}$$

where $D(\rho\|\sigma) = +\infty$ unless $\mathrm{supp}\,\rho \subseteq \mathrm{supp}\,\sigma$, and $\varepsilon \in (0,1)$. For either divergence $\mathbf{D}$ in Eq. (1), its stabilized channel extension is

$$\mathbf{D}_{\mathrm{ch}}(\mathcal{N}\|\mathcal{M}) := \sup_{\psi_{RA} \in \mathcal{D}(R \otimes A)} \mathbf{D}((\mathrm{id}_R \otimes \mathcal{N})(\psi)\|(\mathrm{id}_R \otimes \mathcal{M})(\psi))\,, \qquad R \simeq A. \tag{2}$$

Using Eq. (2), define the regularized channel relative entropy by

$$D_{\mathrm{ch}}^{\infty}(\mathcal{N}\|\mathcal{M}) := \lim_{n\to\infty} \frac{1}{n} D_{\mathrm{ch}}(\mathcal{N}^{\otimes n}\|\mathcal{M}^{\otimes n}). \tag{3}$$

Whenever Eq. (3) is finite, does the fixed-error parallel Stein limit exist and satisfy

$$\lim_{n\to\infty} \frac{1}{n} D_{H,\mathrm{ch}}^{\varepsilon}(\mathcal{N}^{\otimes n}\|\mathcal{M}^{\otimes n}) = D_{\mathrm{ch}}^{\infty}(\mathcal{N}\|\mathcal{M}) \qquad \text{for every } \varepsilon \in (0,1)? \tag{4}$$

Status

Unsolved

Source

Fang, Gour, and Wang prove the weak channel Stein lemma and identify the fixed-error, equivalently strong-converse-threshold, extension as unresolved for general channel pairs [FGW25].

Progress

- The weak channel Stein lemma proves

$$\lim_{\delta\downarrow 0} \lim_{n\to\infty} \frac{1}{n} D_{H,\mathrm{ch}}^{\delta}(\mathcal{N}^{\otimes n}\|\mathcal{M}^{\otimes n}) = D_{\mathrm{ch}}^{\infty}(\mathcal{N}\|\mathcal{M}). \tag{5}$$

  Equation (5) yields the required lower bound at every fixed error, but not the converse inequality [FGW25].
- Sandwiched-Rényi converses bound the fixed-error limsup by

$$\limsup_{n\to\infty} \frac{1}{n} D_{H,\mathrm{ch}}^{\varepsilon}(\mathcal{N}^{\otimes n}\|\mathcal{M}^{\otimes n}) \le \inf_{\alpha>1} \widetilde{D}_{\alpha,\mathrm{ch}}^{\infty}(\mathcal{N}\|\mathcal{M}). \tag{6}$$

  Closing Eq. (6) requires the still-unproved continuity identity

$$\inf_{\alpha>1} \widetilde{D}_{\alpha,\mathrm{ch}}^{\infty}(\mathcal{N}\|\mathcal{M}) \stackrel{?}{=} D_{\mathrm{ch}}^{\infty}(\mathcal{N}\|\mathcal{M}). \tag{7}$$

  Equation (7) is the general strong-converse threshold problem [FGW25].
- The equality in Eq. (4) is known when $\mathcal{M}$ is a replacer channel [CMW16]. It is also known for suitable pairs of idempotent channels that share a full-rank invariant state; this structured result explicitly leaves the general channel case open [SB26].

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Comment

The question concerns parallel tests with an arbitrary entangled input across the channel uses. The weak-error result and the known structured channel families do not establish Eq. (4) for an arbitrary finite-dimensional pair $\mathcal{N}, \mathcal{M}$.

Tag

Quantum hypothesis testing; Channel discrimination; Quantum channels; Quantum information theory

ID

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47 Generalized Stein lemma for fully quantum channel resources

Problem. Let $\mathcal{N} : \mathcal{L}(A) \to \mathcal{L}(B)$ be a finite-dimensional quantum channel. For each n, let $\mathfrak{F}_n$ be a nonempty compact, convex, permutation-invariant set of channels from $A^{\otimes n}$ to $B^{\otimes n}$, closed under tensor products and containing $\mathcal{R}_\omega^{\otimes n}$ for one full-rank state ω, where $\mathcal{R}_\omega(X) := \mathrm{Tr}(X)\omega$. Define

$$D_{\mathrm{ch}}(\mathcal{N}^{\otimes n} \| \mathcal{M}_n) := \sup_{\psi_{R_n A^n}} D\big((\mathrm{id}_{R_n} \otimes \mathcal{N}^{\otimes n})(\psi) \big\| (\mathrm{id}_{R_n} \otimes \mathcal{M}_n)(\psi)\big), \qquad R_n \simeq A^{\otimes n}, \tag{1}$$

where the supremum in Eq. (1) is over density operators and D is the quantum relative entropy. The distance to the free set is

$$E_n^{\mathrm{QQ}}(\mathcal{N} \| \mathfrak{F}_n) := \inf_{\mathcal{M}_n \in \mathfrak{F}_n} D_{\mathrm{ch}}(\mathcal{N}^{\otimes n} \| \mathcal{M}_n). \tag{2}$$

Equation (2) is the n-use relative-entropy distance to the free channel set. For $\varepsilon \in (0,1)$, define the optimal worst-case type-II error of a parallel quantum-input/quantum-output test by

$$\beta_{\varepsilon,n}^{\mathrm{QQ}}(\mathcal{N} \| \mathfrak{F}_n) := \inf_{\substack{\psi_{R_n A^n},\ 0 \le Q \le I \\ \mathrm{Tr}[Q(\mathrm{id}_{R_n} \otimes \mathcal{N}^{\otimes n})(\psi)] \ge 1-\varepsilon}} \ \sup_{\mathcal{M}_n \in \mathfrak{F}_n} \mathrm{Tr}[Q(\mathrm{id}_{R_n} \otimes \mathcal{M}_n)(\psi)]\,. \tag{3}$$

Under what additional structural assumptions on $(\mathfrak{F}_n)_{n \ge 1}$, if any, do both limits exist and obey the fully quantum generalized Stein identity

$$\lim_{n \to \infty} -\frac{1}{n} \log_2 \beta_{\varepsilon,n}^{\mathrm{QQ}}(\mathcal{N} \| \mathfrak{F}_n) = \lim_{n \to \infty} \frac{1}{n} E_n^{\mathrm{QQ}}(\mathcal{N} \| \mathfrak{F}_n) \qquad \text{for every } \varepsilon \in (0,1)? \tag{4}$$

Status

Unsolved

Source

The fully quantum formulation is implicit in the two generalized Stein theorems for classical-input channels, both of which isolate their classical-input structure from the unresolved quantum-input setting [HY25b], [BDK25].

Progress

- For an i.i.d. resource state tested against admissible composite sets of free states, the generalized quantum Stein lemma identifies the fixed-error exponent with the regularized relative entropy of resource [HY25a].
- Two independent works prove the channel analogue for classical–quantum channels. In that setting the channel relative entropy reduces to

$$D_{\mathrm{CQ}}(\Phi \| \Psi) = \max_x D(\rho_x \| \sigma_x), \qquad \Phi : x \mapsto \rho_x, \quad \Psi : x \mapsto \sigma_x, \tag{5}$$

  and its pointwise structure supplies the additivity and minimax steps needed for a fixed-error theorem. These steps apply to Eq. (5) but are unavailable in this form for QQ channels [HY25b], [BDK25].
- The CQ proof explicitly states that analogous properties for fully quantum channels remain unclear. Entangled quantum inputs introduce a reference system and a nontrivial order between input optimization and the worst-case free-channel optimization in Eq. (3) [HY25b].

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Comment

The state theorem and the CQ-channel theorems do not imply Eq. (4) for genuinely quantum inputs. An adaptive quantum-comb version would be a further problem and is not included in the present statement.

Tag

Quantum hypothesis testing; Quantum resource theories; Quantum channels; Superchannels

ID

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48 Square-root remainder in generalized quantum equipartition

Problem. Let H be a d-dimensional Hilbert space. For each n, let $\mathcal{A}_n \subseteq \mathcal{D}(H^{\otimes n})$ and $\mathcal{B}_n \subseteq \mathcal{L}(H^{\otimes n})_+$ be nonempty compact, convex, permutation-invariant sets, each closed under tensor products. For a positive-operator set $\mathcal{C}$, define its positive polar by

$$\mathcal{C}_+^\circ := \{X \geq 0 : \operatorname{Tr}(XY) \leq 1 \text{ for every } Y \in \mathcal{C}\}. \tag{1}$$

Assume that both families of polars from Eq. (1) satisfy

$$(\mathcal{A}_m)_+^\circ \otimes (\mathcal{A}_n)_+^\circ \subseteq (\mathcal{A}_{m+n})_+^\circ, \qquad (\mathcal{B}_m)_+^\circ \otimes (\mathcal{B}_n)_+^\circ \subseteq (\mathcal{B}_{m+n})_+^\circ. \tag{2}$$

In addition to Eq. (2), assume that some constant $C < \infty$ satisfies

$$D_{\max}(\rho_n \| \sigma_n) \leq Cn, \qquad \log_2 \operatorname{Tr} \sigma_n \leq Cn \tag{3}$$

for all $\rho_n \in \mathcal{A}_n$ and $\sigma_n \in \mathcal{B}_n$. The linear growth condition in Eq. (3) uses $D_{\max}(\rho\|\sigma) := \inf\{\lambda : \rho \leq 2^\lambda \sigma\}$. For positive operators ρ and σ, define

$$\begin{aligned} D(\rho\|\sigma) &:= \operatorname{Tr}[\rho(\log_2 \rho - \log_2 \sigma)], \\ D_H^\varepsilon(\rho\|\sigma) &:= -\log_2 \inf_{\substack{0 \leq Q \leq I \\ \operatorname{Tr}(Q\rho) \geq 1-\varepsilon}} \operatorname{Tr}(Q\sigma), \end{aligned} \tag{4}$$

where $D(\rho\|\sigma) = +\infty$ unless $\operatorname{supp} \rho \subseteq \operatorname{supp} \sigma$. Define the divergences between the two sets from Eq. (4) by

$$\begin{aligned} D(\mathcal{A}_n \| \mathcal{B}_n) &:= \inf_{\rho_n \in \mathcal{A}_n,\, \sigma_n \in \mathcal{B}_n} D(\rho_n \| \sigma_n), \\ D_H^\varepsilon(\mathcal{A}_n \| \mathcal{B}_n) &:= \inf_{\rho_n \in \mathcal{A}_n,\, \sigma_n \in \mathcal{B}_n} D_H^\varepsilon(\rho_n \| \sigma_n), \end{aligned} \tag{5}$$

The two quantities in Eq. (5) determine the first- and finite-blocklength orders of interest. Set

$$D^\infty(\mathcal{A}\|\mathcal{B}) := \lim_{n\to\infty} \frac{1}{n} D(\mathcal{A}_n \| \mathcal{B}_n). \tag{6}$$

Does every fixed $\varepsilon \in (0,1)$ admit a constant $K_\varepsilon < \infty$ such that, for all sufficiently large n,

$$|D_H^\varepsilon(\mathcal{A}_n\|\mathcal{B}_n) - nD^\infty(\mathcal{A}\|\mathcal{B})| \leq K_\varepsilon \sqrt{n}? \tag{7}$$

Status

Unsolved

Source

Fang, Fawzi, and Fawzi establish an $O(n^{2/3} \log n)$ generalized equipartition remainder and explicitly leave improvement to the $O(\sqrt{n})$ scale open [FFF26].

Progress

- For two fixed i.i.d. states, quantum hypothesis testing has the second-order expansion

$$D_H^\varepsilon(\rho^{\otimes n} \| \sigma^{\otimes n}) = nD(\rho\|\sigma) + \sqrt{nV(\rho\|\sigma)}\, \Phi^{-1}(\varepsilon) + O(\log n), \tag{8}$$

  where V is the relative-entropy variance and Φ is the standard normal distribution function [Li14].
- Under the assumptions in the problem, the generalized quantum asymptotic equipartition property proves the first-order limit in Eq. (6) for every fixed error [FFF26].
- The best general quantitative estimate currently established is

$$D_H^\varepsilon(\mathcal{A}_n\|\mathcal{B}_n) - nD^\infty(\mathcal{A}\|\mathcal{B}) = O(n^{2/3} \log n) \tag{9}$$

  at fixed ε. The authors explicitly leave replacement of Eq. (9) by the square-root scale in Eq. (7) open [FFF26].

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Comment

Equation (7) asks only for a universal order bound. Identifying a Gaussian coefficient analogous to the variance term in Eq. (8) would be a strictly stronger problem.

Tag

Quantum hypothesis testing; Finite-blocklength quantum information; One-shot quantum information; Quantum information theory

ID

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49 Quantum capacity of a bosonic thermal attenuator

Problem. Let $\Phi_{\eta,\nu}$ be the single-mode bosonic thermal attenuator with transmissivity $0 < \eta < 1$ and environmental mean photon number $0 < \nu < \infty$, defined by

$$\Phi_{\eta,\nu}(\rho_A) := \mathrm{Tr}_{E'}\left[U_\eta(\rho_A \otimes \tau_{\nu,E})U_\eta^\dagger\right], \qquad \tau_\nu := \sum_{k=0}^{\infty} \frac{\nu^k}{(\nu+1)^{k+1}}|k\rangle\langle k|, \tag{1}$$

where $U_\eta : AE \to BE'$ is a beam-splitter unitary. Its unassisted quantum capacity is the regularized coherent information

$$\mathcal{Q}(\Phi_{\eta,\nu}) := \lim_{n\to\infty} \frac{1}{n} \sup_{\rho_{A^n}} I_c(\rho_{A^n}, \Phi_{\eta,\nu}^{\otimes n}), \qquad I_c(\rho, \mathcal{N}) := S(\mathcal{N}(\rho)) - S(\mathcal{N}^c(\rho)), \tag{2}$$

where $S(\sigma) := -\mathrm{Tr}(\sigma \log_2 \sigma)$ and $\mathcal{N}^c$ is any complementary channel. Determine Eq. (2) for all finite-temperature parameters in Eq. (1).

Status

Unsolved

Source

Rosati, Mari, and Giovannetti explicitly study the unknown thermal-attenuator quantum capacity through nonmatching narrow bounds; the exact-capacity problem is retained after the later non-Gaussian separation [RMG18], [MCF+26].

Progress

- At zero environmental temperature, the pure-loss capacity is

$$\mathcal{Q}(\Phi_{\eta,0}) = \left[\log_2 \frac{\eta}{1-\eta}\right]_+, \qquad [x]_+ := \max\{0, x\}. \tag{3}$$

  Thus Eq. (3) solves only the boundary case $\nu = 0$ [WPG07].

- Optimizing the one-use coherent information over all single-mode Gaussian input states gives exactly

$$\sup_{\rho\in\mathsf{G}_1} I_c(\rho, \Phi_{\eta,\nu}) = \left[\log_2 \frac{\eta}{1-\eta} - g(\nu)\right]_+, \qquad g(x) := (x+1)\log_2(x+1) - x\log_2 x. \tag{4}$$

  The rate in Eq. (4) was first obtained from thermal inputs, but it is not the optimum over arbitrary non-Gaussian inputs [HW01], [MCF+26].

- The exact antidegradability region is

$$\eta \le \eta_{\mathrm{AD}}(\nu) := \frac{\nu + \frac{1}{2}}{\nu+1}, \qquad \mathcal{Q}(\Phi_{\eta,\nu}) = 0. \tag{5}$$

  Hence Eq. (2) is settled throughout the region in Eq. (5) [LKA+19].

- A bottleneck decomposition supplies a generally nonmatching upper bound:

$$\mathcal{Q}(\Phi_{\eta,\nu}) \le \left[\log_2 \frac{\eta - (1-\eta)\nu}{(1-\eta)(\nu+1)}\right]_+ \quad \text{when } \eta > (1-\eta)\nu, \tag{6}$$

  while the channel is entanglement breaking and has zero capacity otherwise [RMG18]. The bound in Eq. (6) does not generally coincide with the Gaussian achievable rate.

- Non-Gaussian inputs strictly exceed the Gaussian value. In particular,

$$\sup_{\rho\in\mathsf{G}_1} I_c(\rho, \Phi_{4/5,1}) = 0, \qquad \mathcal{Q}(\Phi_{4/5,1}) \ge 4.7 \times 10^{-4}. \tag{7}$$

  Equation (7) rules out a restriction to single-mode Gaussian inputs but does not determine the capacity [MCF+26].

References

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Comment

For generic $\nu > 0$ outside the zero-capacity regions, the lower and upper bounds do not coincide. Non-Gaussian one-mode optimization and possible multimode superadditivity both remain relevant to Eq. (2).

Tag

Bosonic channels; Quantum capacity; Coherent information; Gaussian quantum information; Continuous-variable systems

ID

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50 Positivity threshold for thermal-attenuator quantum capacity

Problem. For $0 < \eta < 1$ and $0 < \nu < \infty$, let the single-mode bosonic thermal attenuator be

$$\Phi_{\eta,\nu}(\rho_A) := \mathrm{Tr}_{E'}\Big[U_\eta(\rho_A \otimes \tau_{\nu,E})U_\eta^\dagger\Big], \qquad \tau_\nu := \sum_{k=0}^{\infty} \frac{\nu^k}{(\nu+1)^{k+1}} |k\rangle\langle k|, \tag{1}$$

where U_η is a beam-splitter unitary. Define its unassisted quantum capacity and its critical transmissivity by

$$\mathcal{Q}(\Phi_{\eta,\nu}) := \lim_{n\to\infty} \frac{1}{n} \sup_{\rho_{A^n}} I_c(\rho_{A^n}, \Phi_{\eta,\nu}^{\otimes n}), \qquad \eta_c(\nu) := \inf\{\eta \in (0,1) : \mathcal{Q}(\Phi_{\eta,\nu}) > 0\}, \tag{2}$$

where $I_c(\rho, \mathcal{N}) := S(\mathcal{N}(\rho)) - S(\mathcal{N}^c(\rho))$. Determine the exact threshold curve in Eq. (2) for the channel in Eq. (1).

Status

Unsolved

Source

The exact positivity threshold is implicit in the gap between the antidegradability boundary of Lami et al. and the non-Gaussian positive-rate witnesses of Mele et al. [LKA+19], [MCF+26].

Progress

- Exact antidegradability gives the lower bound

$$\eta_c(\nu) \geq \eta_{\mathrm{AD}}(\nu) := \frac{\nu + \frac{1}{2}}{\nu + 1}. \tag{3}$$

  The capacity vanishes at and below the right-hand side of Eq. (3) [LKA+19].

- Thermal input states give the upper bound

$$\eta_c(\nu) \leq \eta_{\mathrm{G}}(\nu) := \frac{2^{g(\nu)}}{1 + 2^{g(\nu)}}, \qquad g(x) := (x+1)\log_2(x+1) - x\log_2 x. \tag{4}$$

  Above the right-hand side of Eq. (4), the thermal-state coherent information is positive [HW01].

- For one environmental thermal photon, certified non-Gaussian inputs and antidegradability narrow the threshold to

$$\frac{3}{4} \leq \eta_c(1) \leq 0.7841 < \eta_{\mathrm{G}}(1) = \frac{4}{5}. \tag{5}$$

  In particular, Eq. (5) proves that the Gaussian threshold is not the true positivity threshold [MCF+26].

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Comment

This is the positivity-boundary subproblem of determining the full capacity in Problem 49. Even the single value in Eq. (5) is not known exactly.

Tag

Bosonic channels; Quantum capacity; Antidegradable channels; Coherent information; Continuous-variable systems

ID

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51 Energy-constrained quantum capacity of a thermal attenuator

Problem. For $0 < \eta < 1$ and $0 < \nu < \infty$, let the single-mode bosonic thermal attenuator be

$$\Phi_{\eta,\nu}(\rho_A) := \mathrm{Tr}_{E'}\left[U_\eta(\rho_A \otimes \tau_{\nu,E})U_\eta^\dagger\right], \qquad \tau_\nu := \sum_{k=0}^{\infty} \frac{\nu^k}{(\nu+1)^{k+1}} |k\rangle\langle k|, \tag{1}$$

where U_η is a beam-splitter unitary. For a finite mean input photon number $0 < N_\mathrm{S} < \infty$, define the energy-constrained unassisted quantum capacity by

$$\mathcal{Q}(\Phi_{\eta,\nu}, N_\mathrm{S}) := \lim_{n\to\infty} \frac{1}{n} \sup_{\substack{\rho_{A^n}: \\ \mathrm{Tr}[\rho_{A^n} \sum_{j=1}^n \hat{n}_j] \le nN_\mathrm{S}}} I_\mathrm{c}(\rho_{A^n}, \Phi_{\eta,\nu}^{\otimes n}), \tag{2}$$

where $\hat{n}_j$ is the photon-number operator of the jth mode and $I_\mathrm{c}(\rho, \mathcal{N}) := S(\mathcal{N}(\rho)) - S(\mathcal{N}^\mathrm{c}(\rho))$. Determine Eq. (2) for the thermal channel in Eq. (1).

Status

Unsolved

Source

The problem is implicit in the nonmatching energy-constrained bounds of Sharma et al. and the strict multimode achievable-rate improvements of Noh, Pirandola, and Jiang [SWA+18], [NPJ20].

Progress

- For the zero-temperature pure-loss channel, the constrained capacity is known exactly:

$$\mathcal{Q}(\Phi_{\eta,0}, N_\mathrm{S}) = [g(\eta N_\mathrm{S}) - g((1-\eta)N_\mathrm{S})]_+, \qquad g(x) := (x+1)\log_2(x+1) - x\log_2 x. \tag{3}$$

  Equation (3) does not extend directly to a thermal environment [WQ18].

- For $\nu > 0$, a thermal input of mean photon number N_S gives

$$\mathcal{Q}(\Phi_{\eta,\nu}, N_\mathrm{S}) \ge [L(\eta,\nu,N_\mathrm{S})]_+, \tag{4}$$

  where the rate in Eq. (4) is

$$\begin{aligned} L(\eta,\nu,N_\mathrm{S}) &:= g(\eta N_\mathrm{S} + (1-\eta)\nu) \\ &\quad - g\left(\frac{\Delta + (1-\eta)N_\mathrm{S} - (1-\eta)\nu - 1}{2}\right) \\ &\quad - g\left(\frac{\Delta - (1-\eta)N_\mathrm{S} + (1-\eta)\nu - 1}{2}\right), \\ \Delta &:= \sqrt{[(1+\eta)N_\mathrm{S} + (1-\eta)\nu + 1]^2 - 4\eta N_\mathrm{S}(N_\mathrm{S}+1)}. \end{aligned} \tag{5}$$

  The expression in Eq. (5) is a one-use achievable rate, not an exact capacity formula [HW01], [SWA+18].

- Multimode Gaussian constructions improve the raw thermal-input rate through the convexified bound

$$\mathcal{Q}(\Phi_{\eta,\nu}, N_\mathrm{S}) \ge \sup_{0<x\le 1} x I_\mathrm{c}(\tau_{N_\mathrm{S}/x}, \Phi_{\eta,\nu}) \ge [L(\eta,\nu,N_\mathrm{S})]_+. \tag{6}$$

  For $\nu = N_\mathrm{S} = 1$, this construction numerically improves the single-mode thermal rate in a nontrivial loss regime [NPJ20].

- A data-processing decomposition gives, for $1/2 \le \eta < 1$,

$$\mathcal{Q}(\Phi_{\eta,\nu}, N_\mathrm{S}) \le [U(\eta,\nu,N_\mathrm{S})]_+, \qquad U := g(\eta' N_\mathrm{S}) - g((1-\eta')N_\mathrm{S}), \quad \eta' := \frac{\eta}{1+(1-\eta)\nu}. \tag{7}$$

  The upper bound in Eq. (7) generally does not meet Eq. (6) [SWA+18].

- The unconstrained optimization over all single-mode Gaussian inputs is known, but its proof fixes output entropy rather than input energy and explicitly does not establish thermal-state optimality at fixed N_S [MCF+26].

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Comment

For finite $\nu > 0$ and finite N_{S}, neither the optimal one-use input nor the necessity of regularization is known. This constrained problem is distinct from the unconstrained capacity in Problem 49.

Tag

Bosonic channels; Quantum capacity; Coherent information; Gaussian quantum information; Continuous-variable systems

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